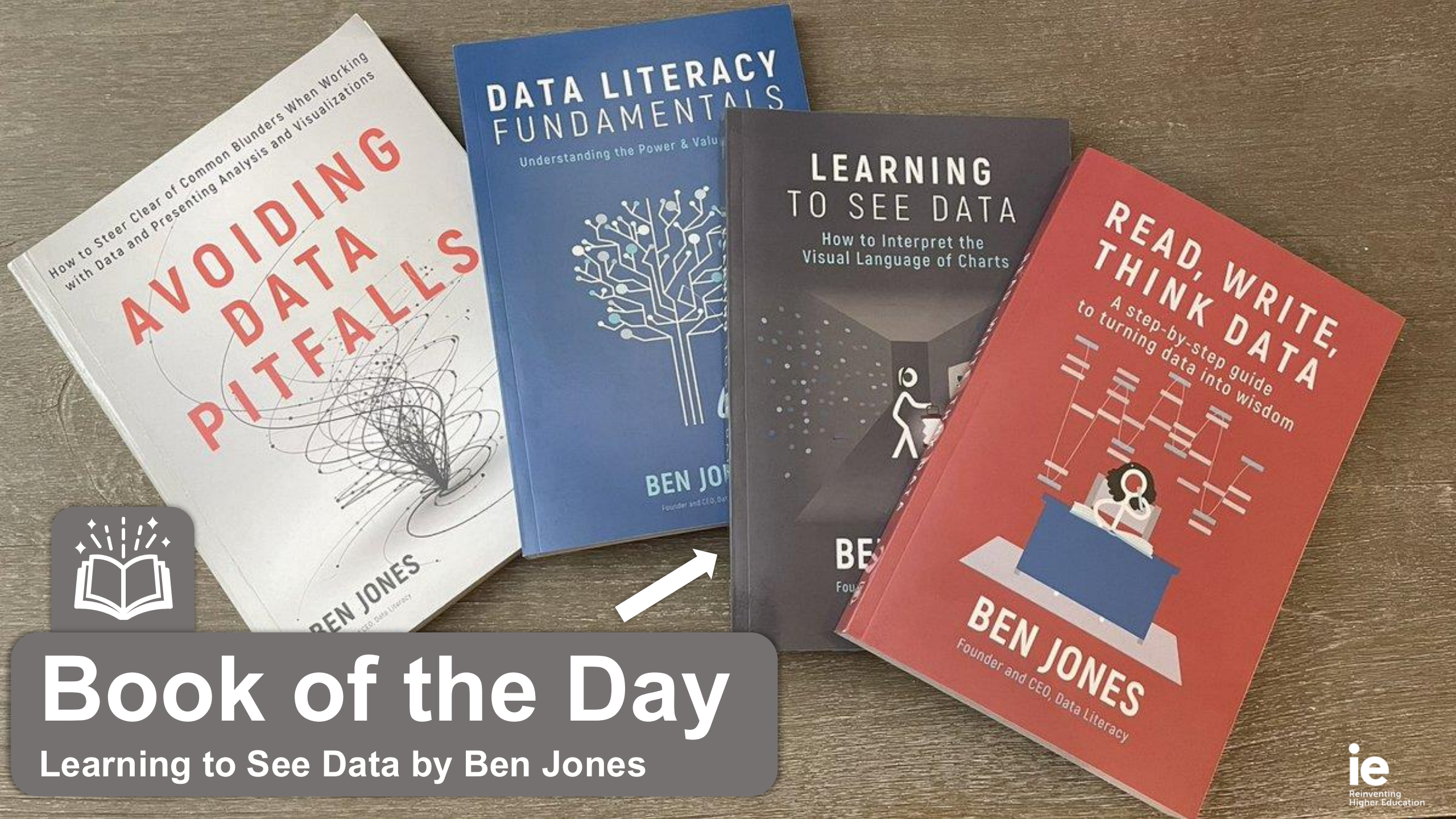
ISA Data Fluency

IMBA Session 2

"Data visualization is the language of decision-making. Good charts effectively convey information. Great charts enable, inform and improve decision-making." –Dante Vitagliano

Professor Christina Stathopoulos, September 2026

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Book of the Day

Learning to See Data by Ben Jones

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O'REILLY®

This Week in AI

Tune in live every Monday at 4pm CET. Or wait to check out the recap article and YouTube recording published on Fridays!

official show link
YouTube playlist

Click each of these images for the episode recap →













Session 2 Agenda



Intro to Data Viz



Design Principles



Best Practices & Common Pitfalls

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Intro to Data Viz

What & Why

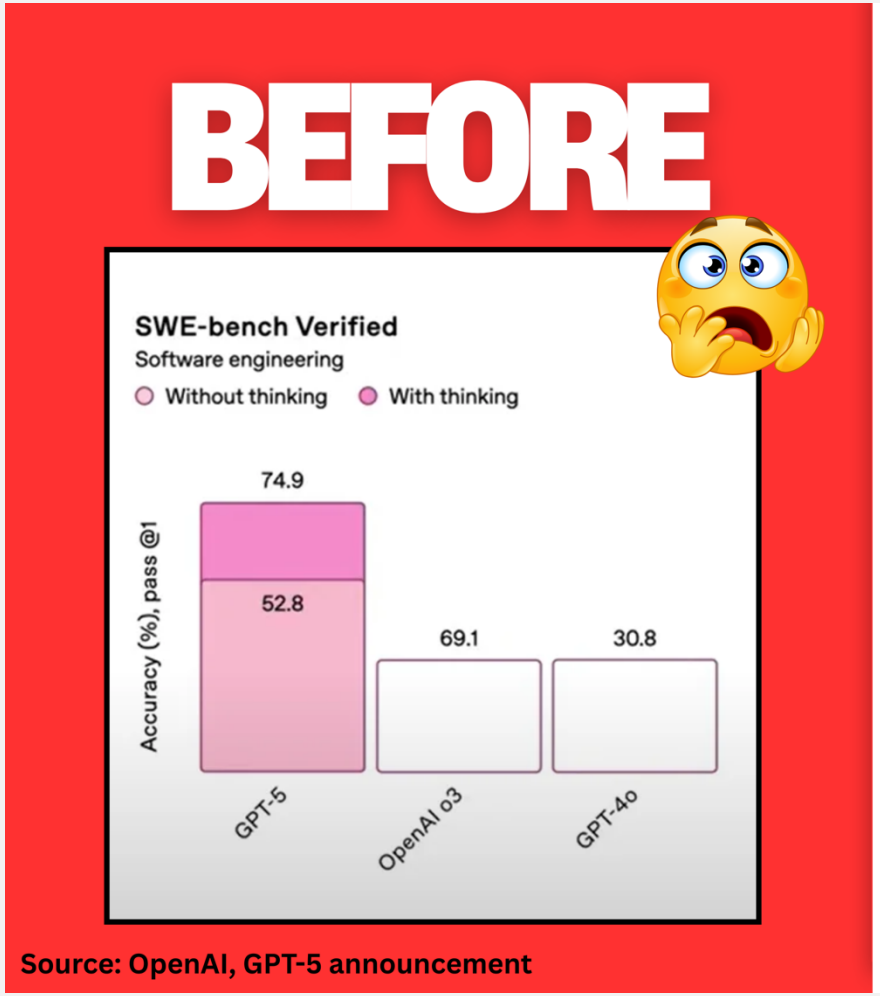


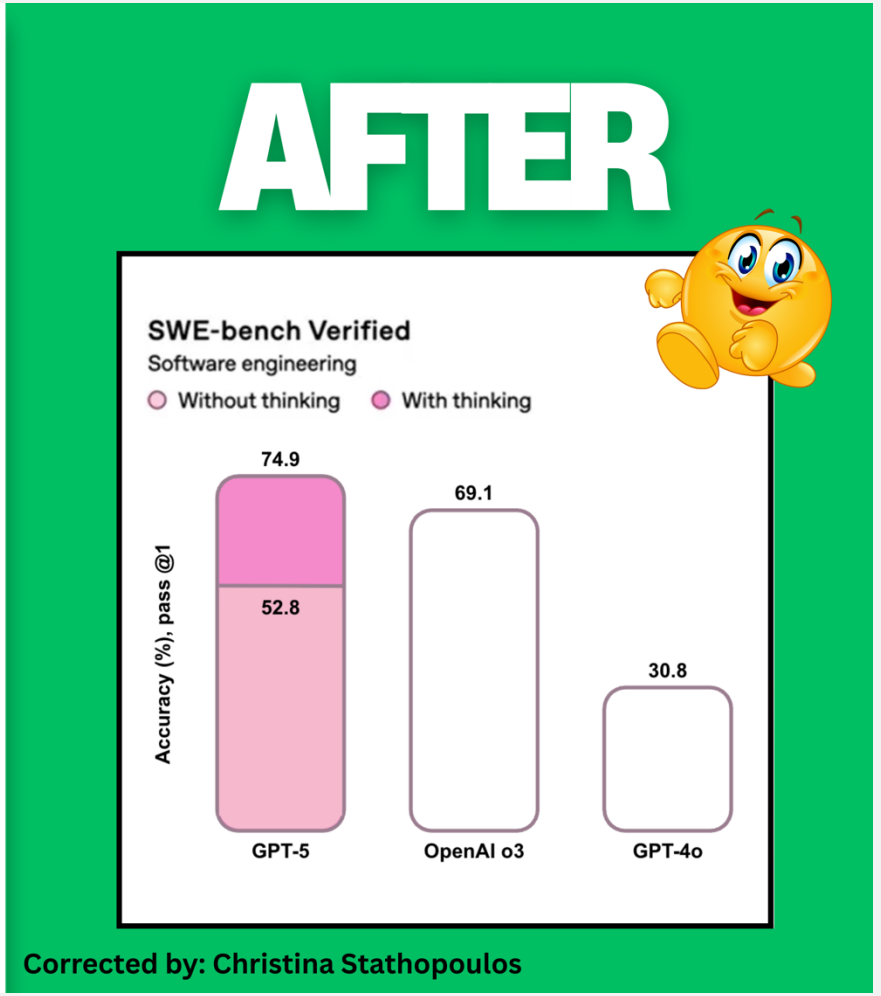
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OpenAI's announcement of GPT-5

August 2025

They have an incredible AI team, and yet they seem to lack the basic fundamental knowledge of data visualization





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Can you figure out what is wrong with this data visualization?



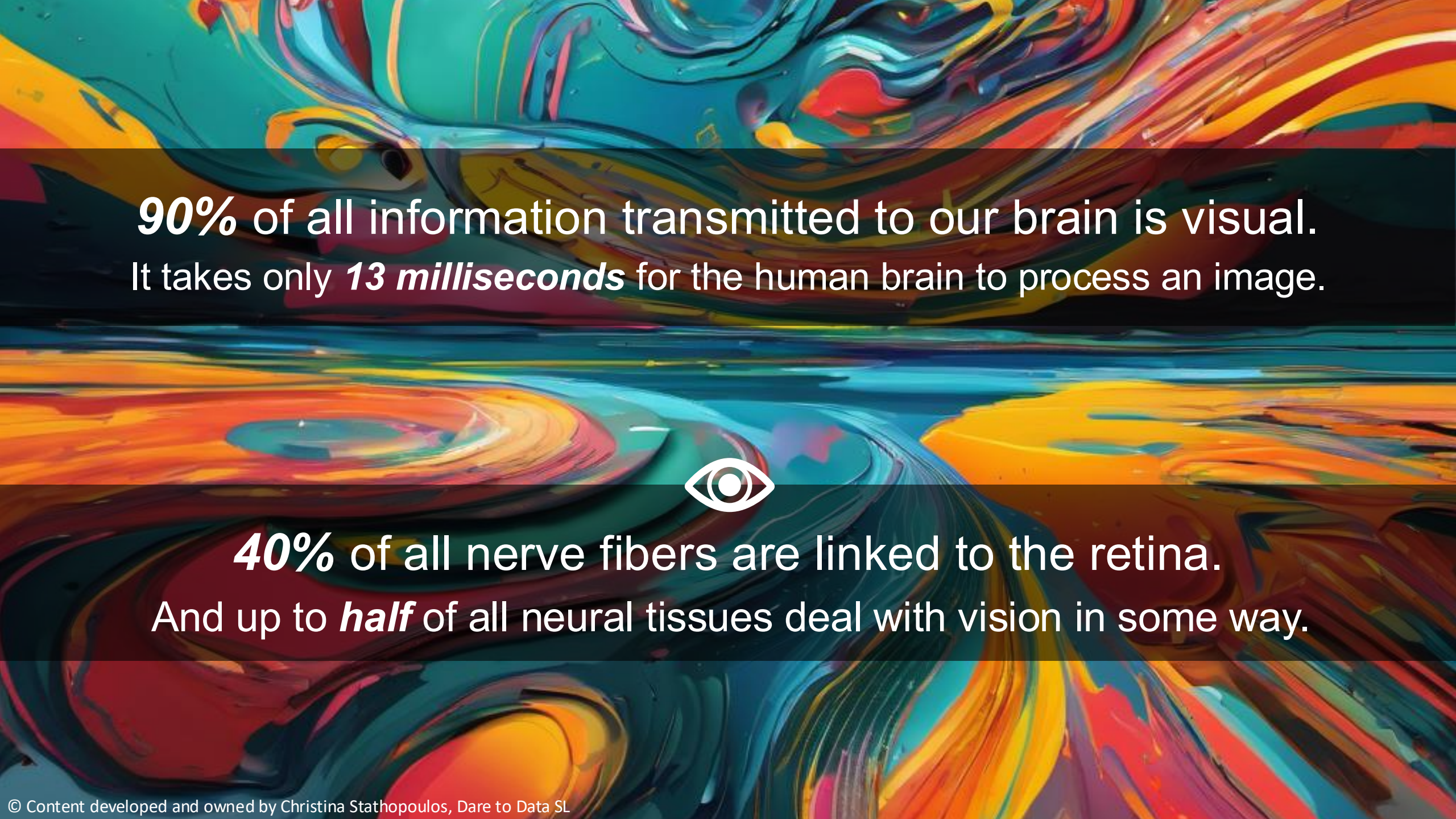
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“Where for most of history we have suffered from a shortage of information, tomorrow we will struggle with a surfeit. Already today, the chief difficulty is not so much obtaining information but **finding the relevant data**.”

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Source: The Economist (2012), Megachange: the World in 2050



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How many F's are there?

HGJALQMDIFGKTPSNAKPENFVL
PQOEPJGAMAFYPQOEYCNMLCA
FHGURNQLFUNVMALQPTUEMSIF
GNCKAPKFNSLEITYGMSLAOQEC
ZXLSPFIGMGBECBAKPQMDKELTUI

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How many F's are there?

HGJALQMDI**F**GKTPSNAKPEN**F**VL
PQOEPJGAMA**F**YPQOEYCNMLCA
FHGURNQL**F**UNVMALQPTUEMSI**F**
GNCKAPK**F**NSLEITYGMSLAOQEC
ZXLSP**F**IGMGBECBAKPQMDKELTUI

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WHAT IS DATA VISUALIZATION?



WHAT IS DATA VISUALIZATION?

Graphical representation of information.

HUGE amounts of data can be easily digestible when represented visually.

STATIC

Tells a specific story & focuses on a specific data relationship.
(e.g. infographic)

INTERACTIVE

Allows users to select specific data points & alter the findings.
(e.g. dashboard)

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Data Visualization Tools

Data Studio: datastudio.google.com

Tableau Public: public.tableau.com

PowerBI: powerbi.microsoft.com

Carto: carto.com

Qlik: qlik.com

D3: d3js.org

Flourish: flourish.studio

Rstudio & ggplot2 package: rstudio.com

Python & packages like Bokeh: continuum.io/downloads

Google Sheets

Excel

Canva

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Python & packages like Bokeh: continuum.io/downloads

Google Sheets

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Design Principles I

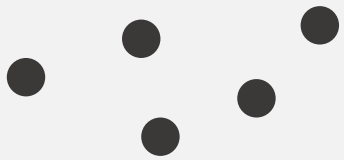
Graphical Marks & Encoding Channels

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Marks

Marks are the **basic visual objects** that represent data.

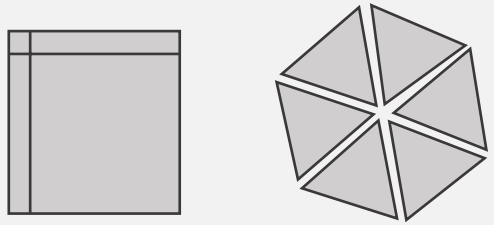
Points 0D



Lines 1D



Areas 2D



(+ Volume) 3D

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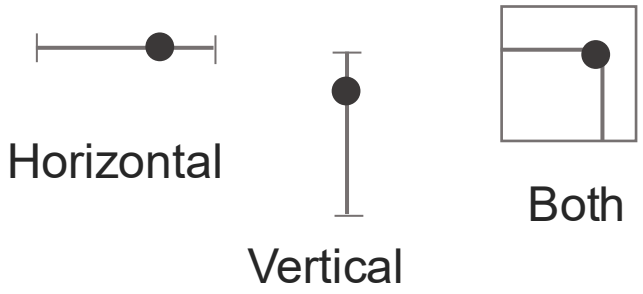
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Source: Munzner, Tamara (2014). Visualization Analysis and Design. CRC Press

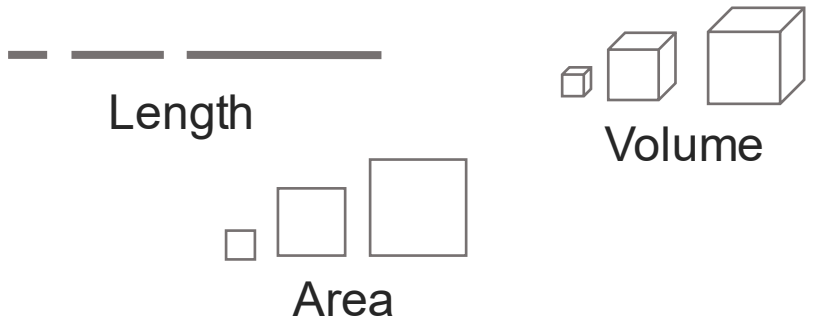
Channels

Channels are **visual variables** we can use to **represent characteristics** of these objects.

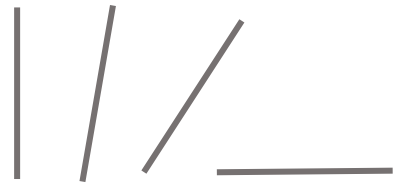
Position



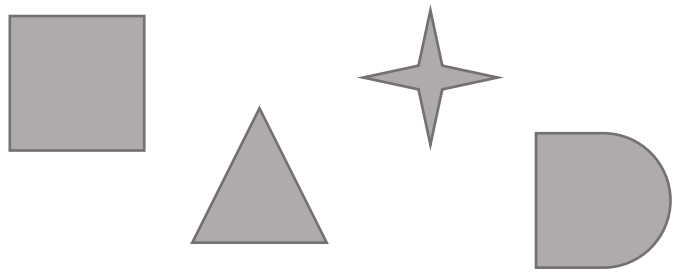
Size



Tilt



Shape



Color



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Design Principles II

Pre-Attentive Attributes

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INTRO

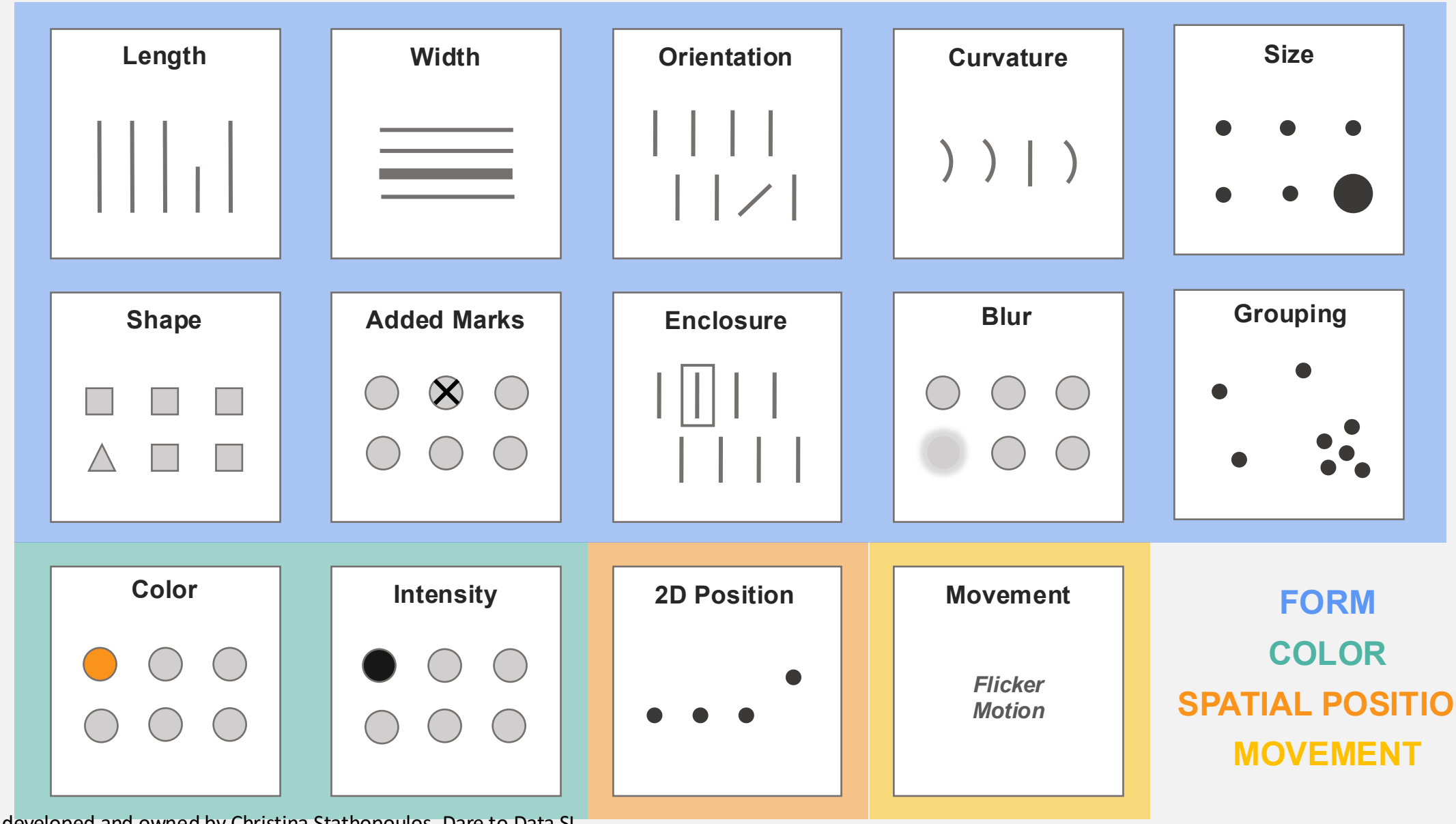


PRE-ATTENTIVE ATTRIBUTES



Pre-attentive Attributes

(Pattern Recognition)

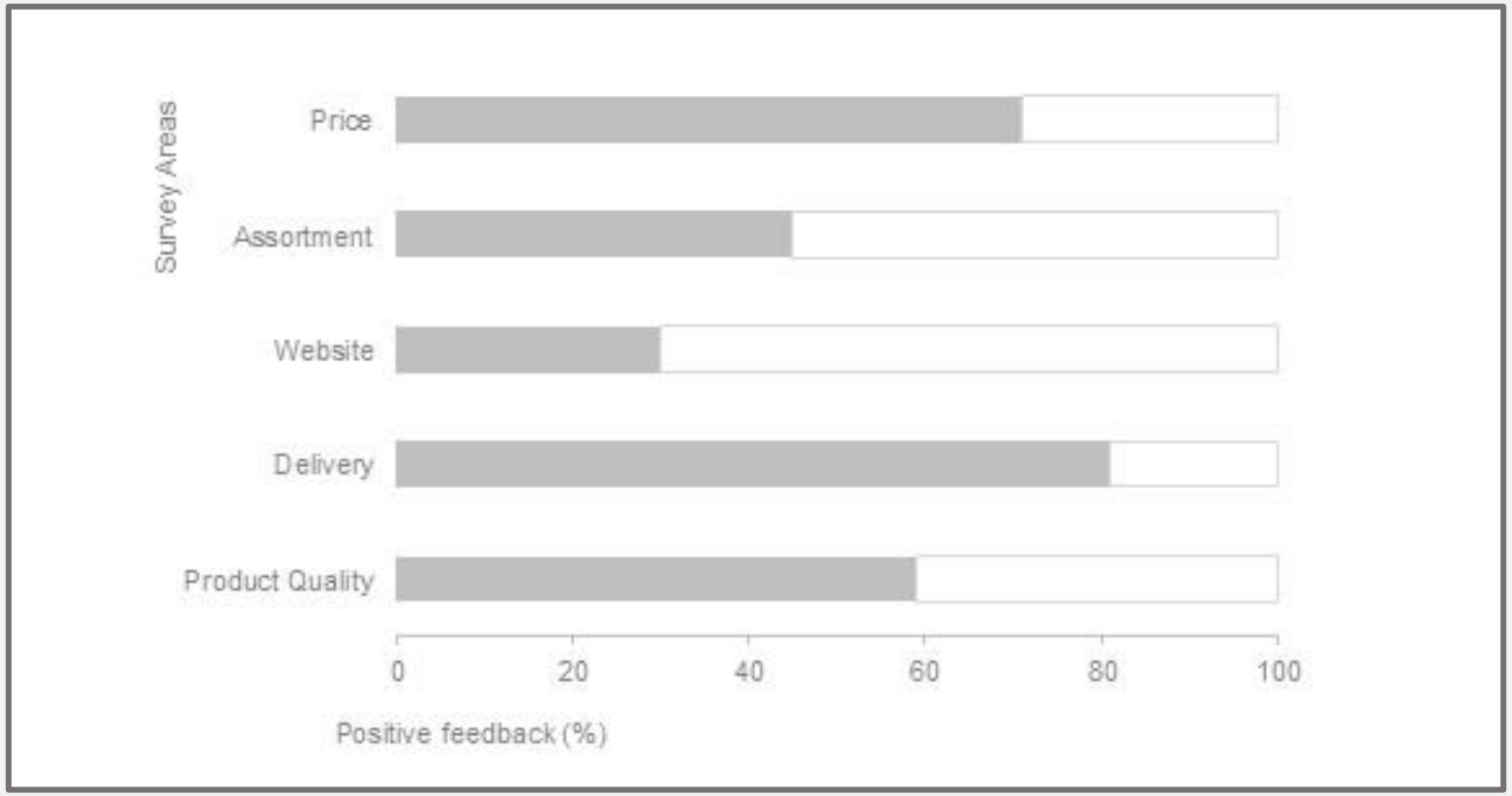


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Source: Preattentive Attributes in Visualization – An example (2016). Daydreaming Numbers

Example – Length

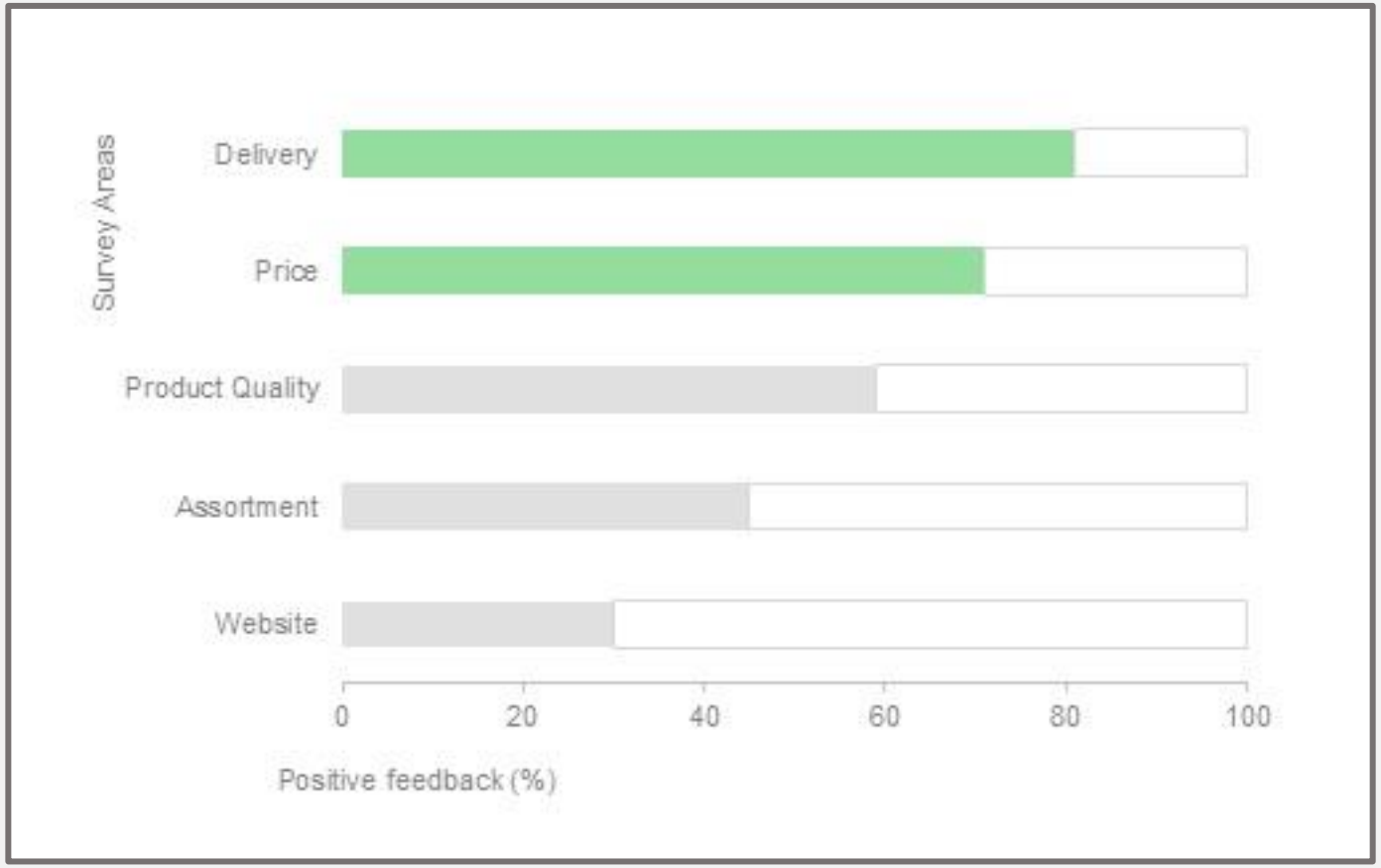


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Source: Preattentive Attributes in Visualization – An example (2016). Daydreaming Numbers

Example – Color

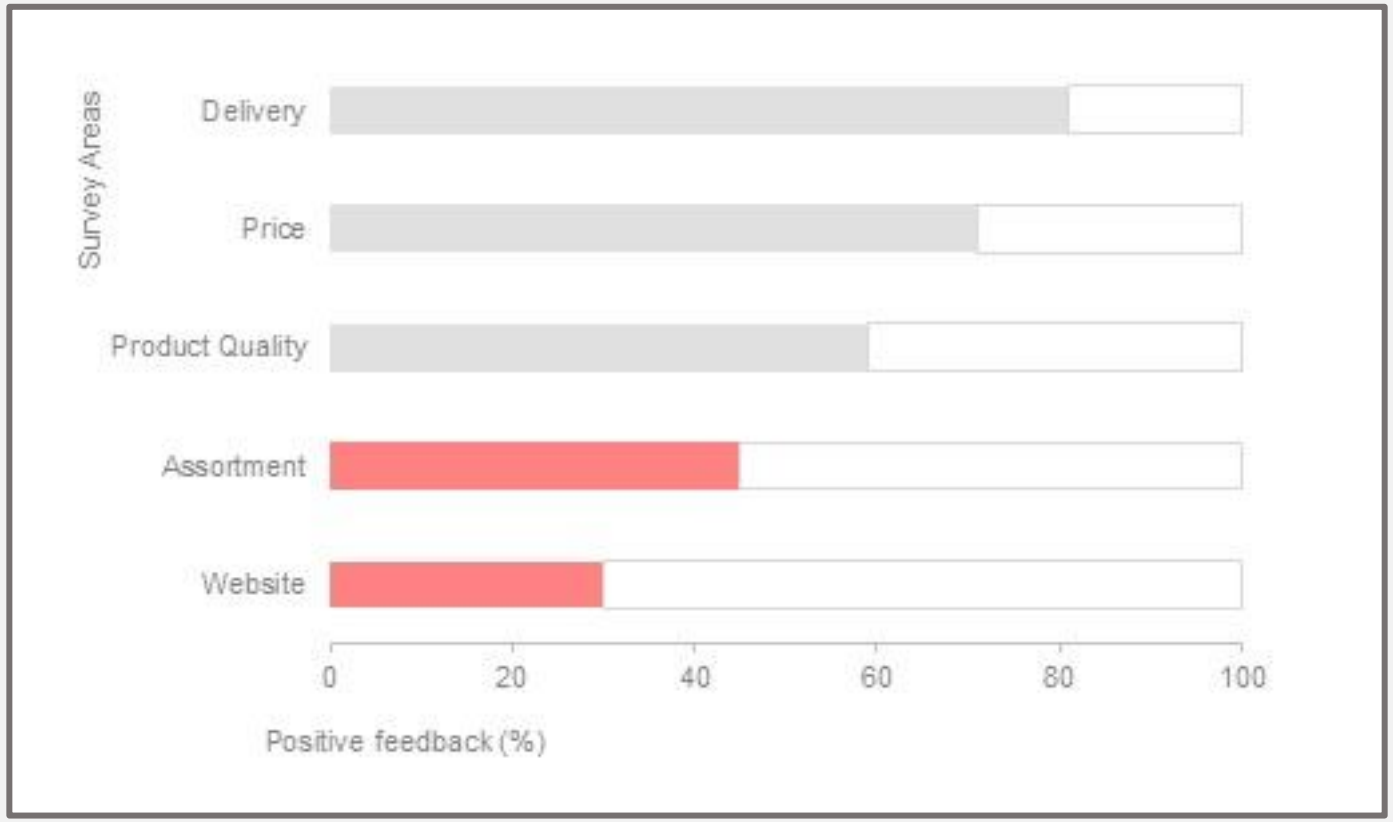


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Source: Preattentive Attributes in Visualization – An example (2016). Daydreaming Numbers

Example – Color

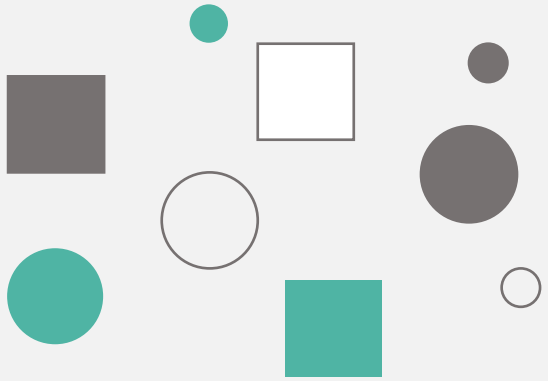


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! A word of warning with pre-attentive attributes !

The more you emphasize **more things**, the **less** your emphasis counts.



Don’t overdo it!

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(homework review)

Design Principles III

Gestalt Principles

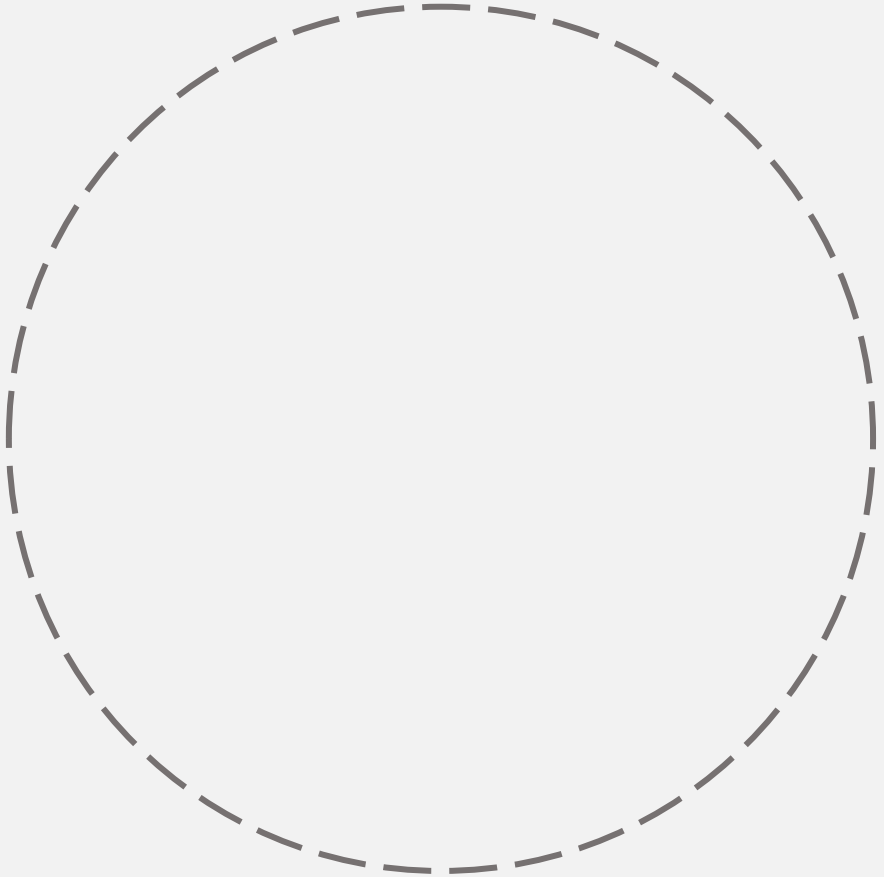


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Our brains like to group things

It's in our nature to simplify things to reduce the cognitive effort to process them.

Do you see this circle? →

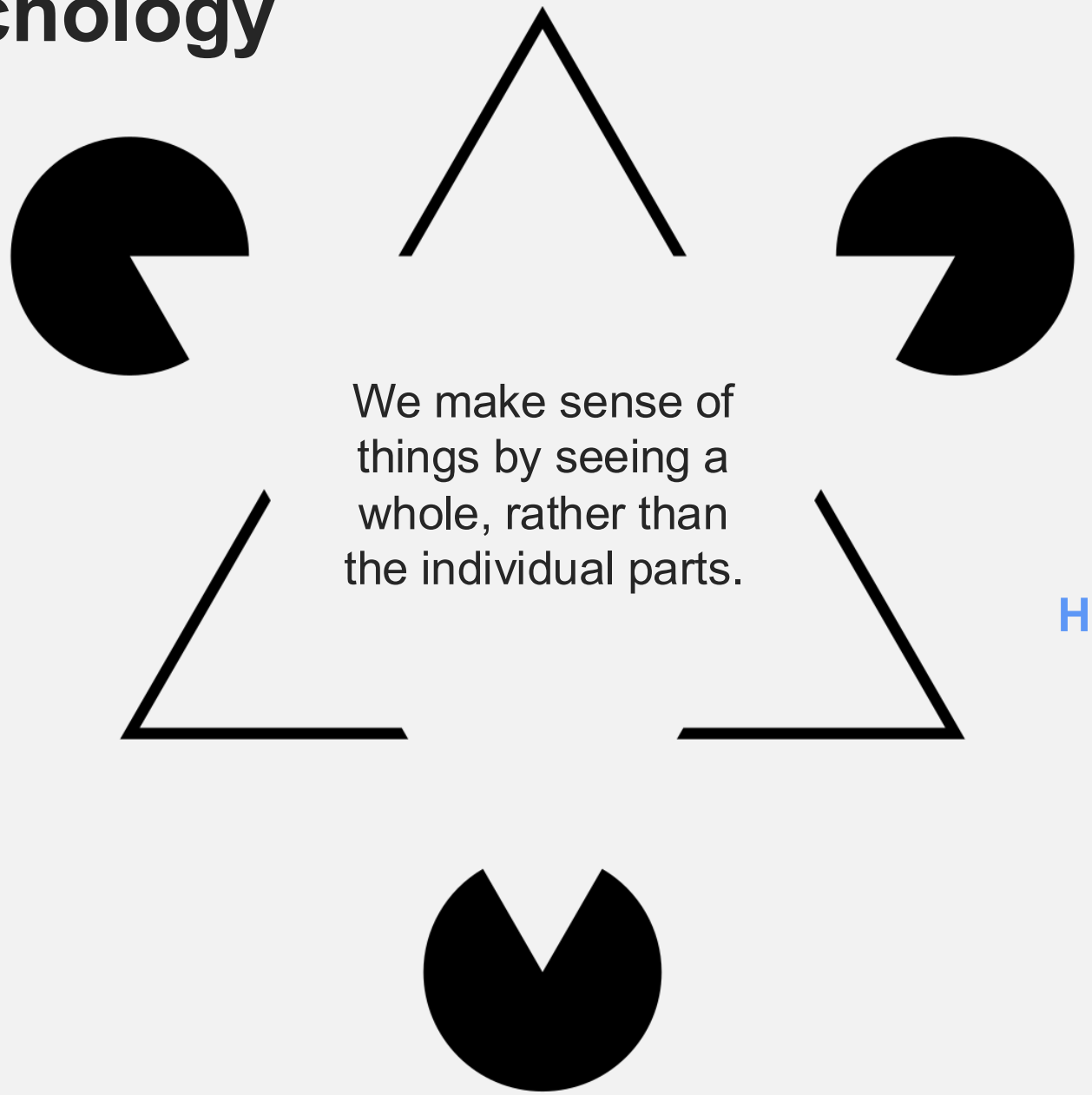


…technically it isn't a circle, but we group the pattern together to form one! Why?....

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Gestalt Psychology



We make sense of things by seeing a whole, rather than the individual parts.

How many shapes do you see here?

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GESTALT PRINCIPLES



ART with KUNSTLER

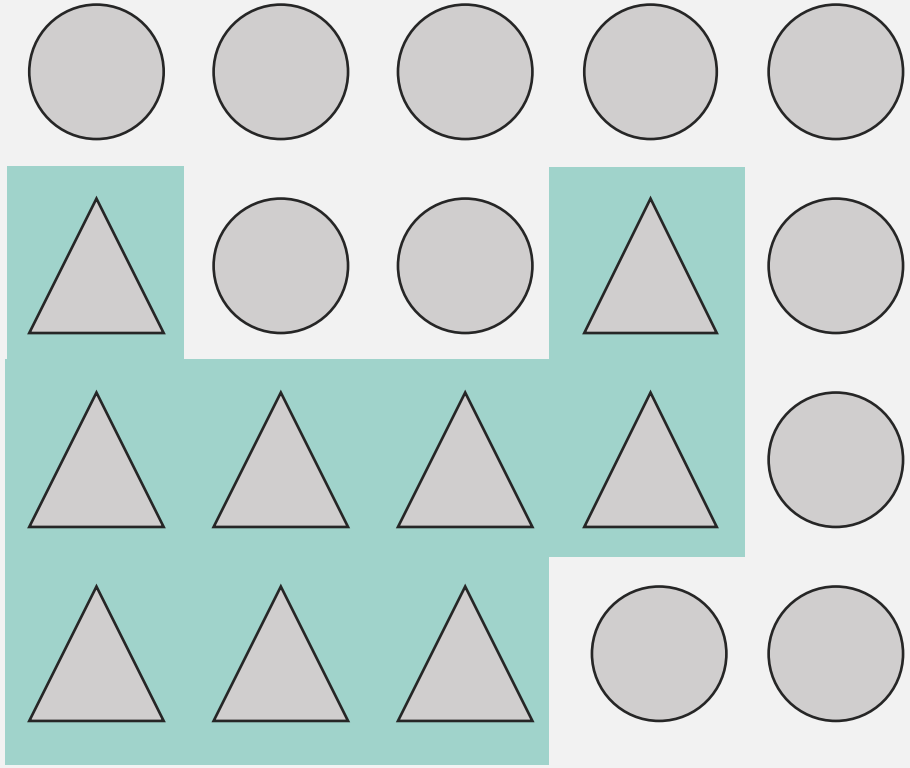
Gestalt Principles

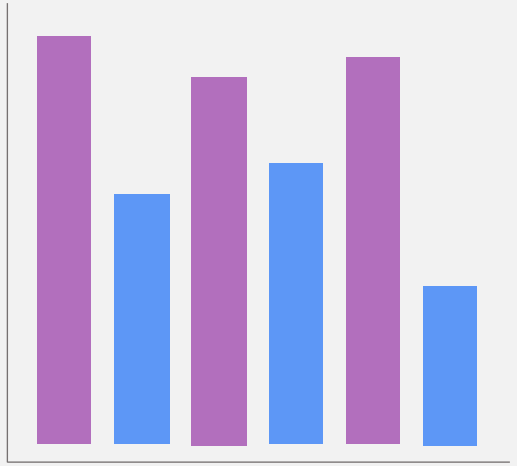
Similarity	Items similar to each other
Proximity	Objects near each other
Common Region	Objects within the same enclosure
Continuity	Carrying on a pattern
Closure	Filling in missing pieces to create a whole
Figure-Ground	Objects in foreground (prominent) or background
Focal Point	Visual piece to capture attention first
Order & Symmetry	Symmetrical elements create a harmonious whole

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Gestalt Principles ▶ Similarity



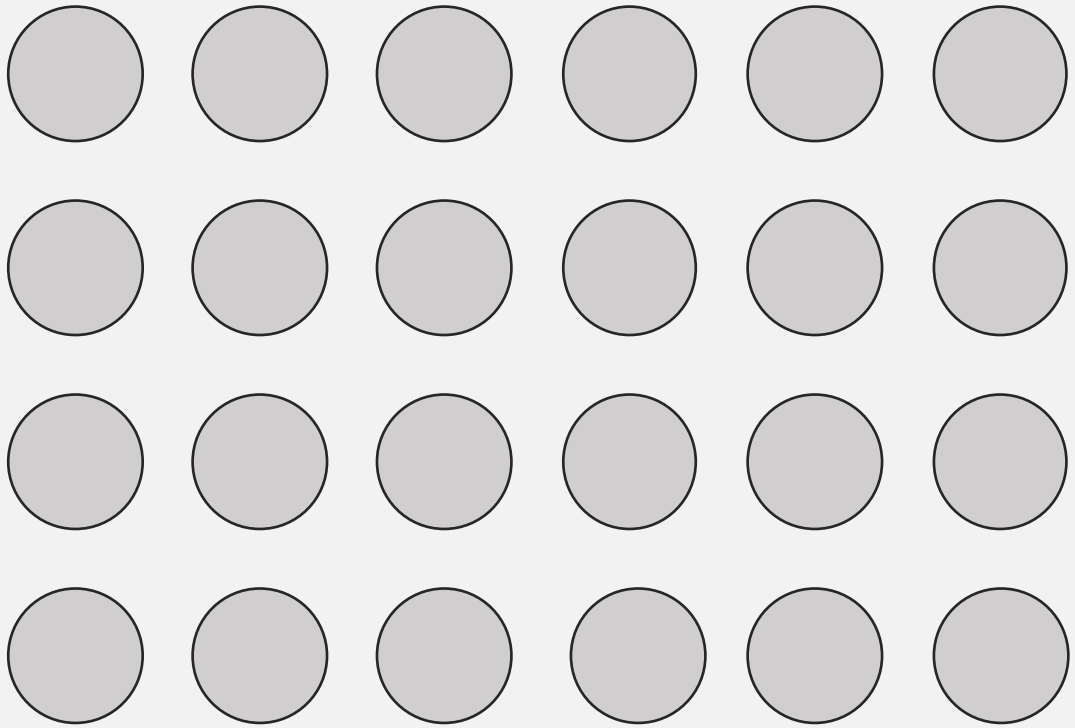


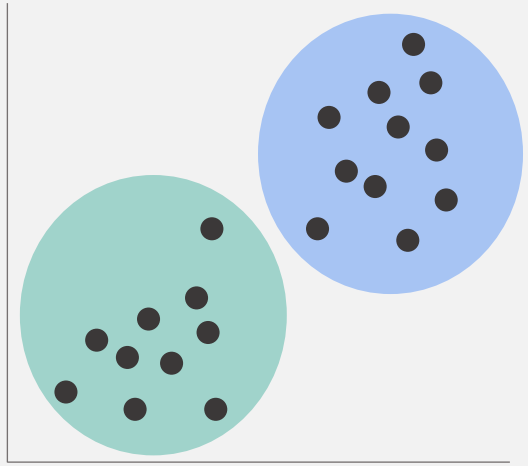
Items that are similar tend to be grouped together.
'Similarity' can be achieved through color, shape, size, etc.

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Gestalt Principles ▶ Proximity





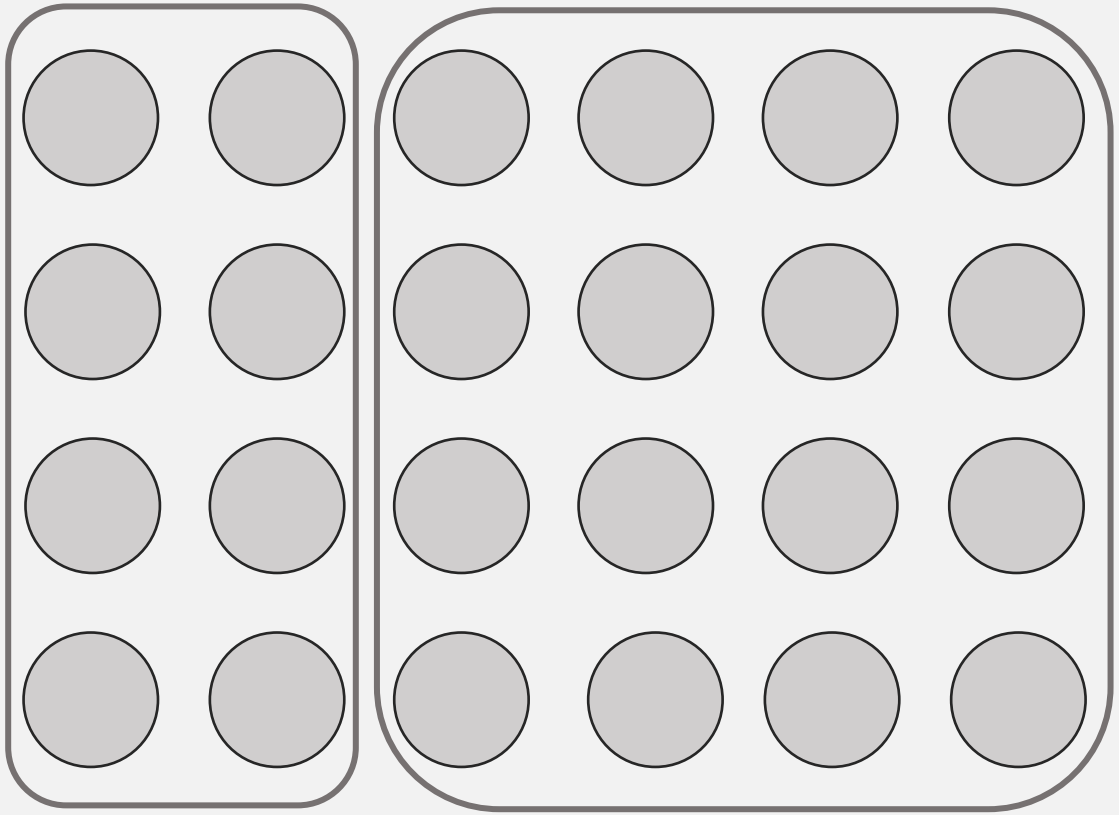
We group data points that seem near each other & assume they share similar characteristics.

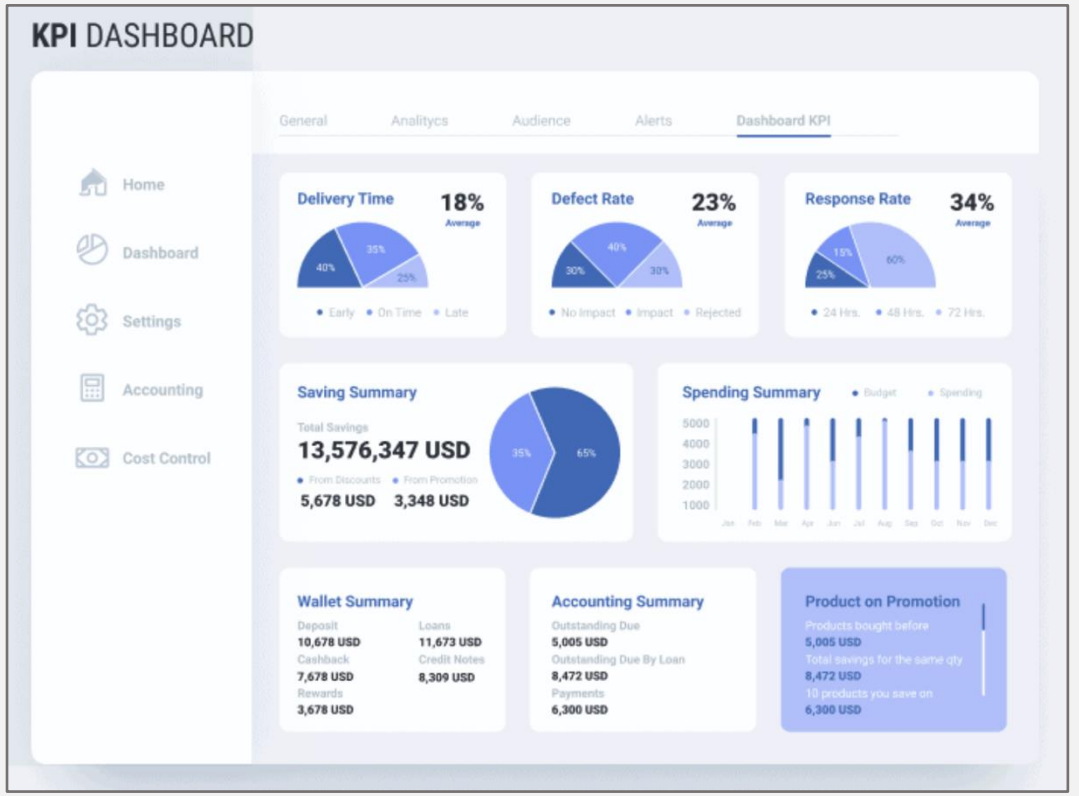
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Gestalt Principles ▶ Common Region

(Enclosure)



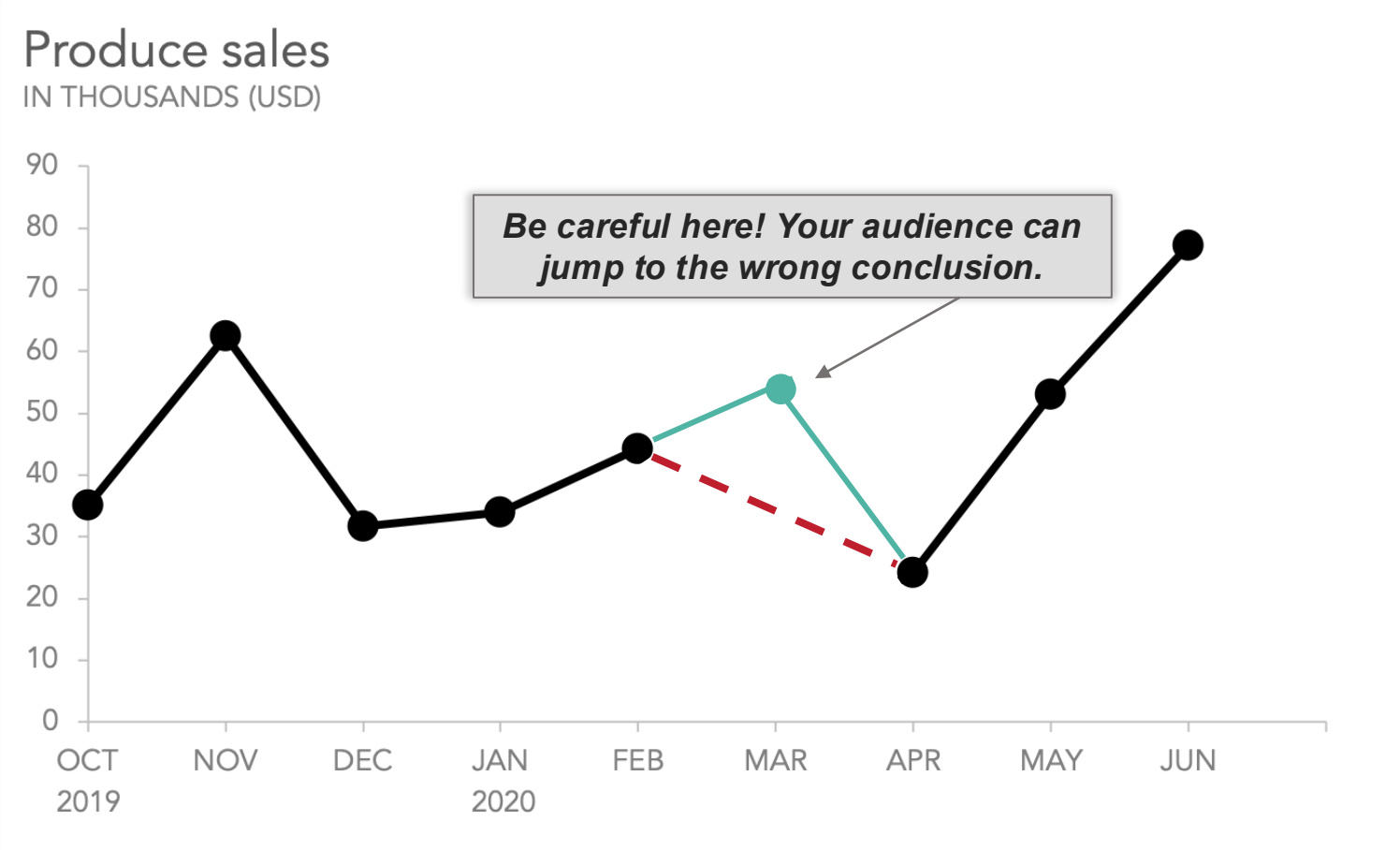


We imagine everything inside a drawn boundary as belonging to the same group.

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Gestalt Principles ▶ Continuity



We have a visual tendency to create continuous smooth paths, essentially 'connecting the dots'.

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Gestalt Principles ▶ Continuity





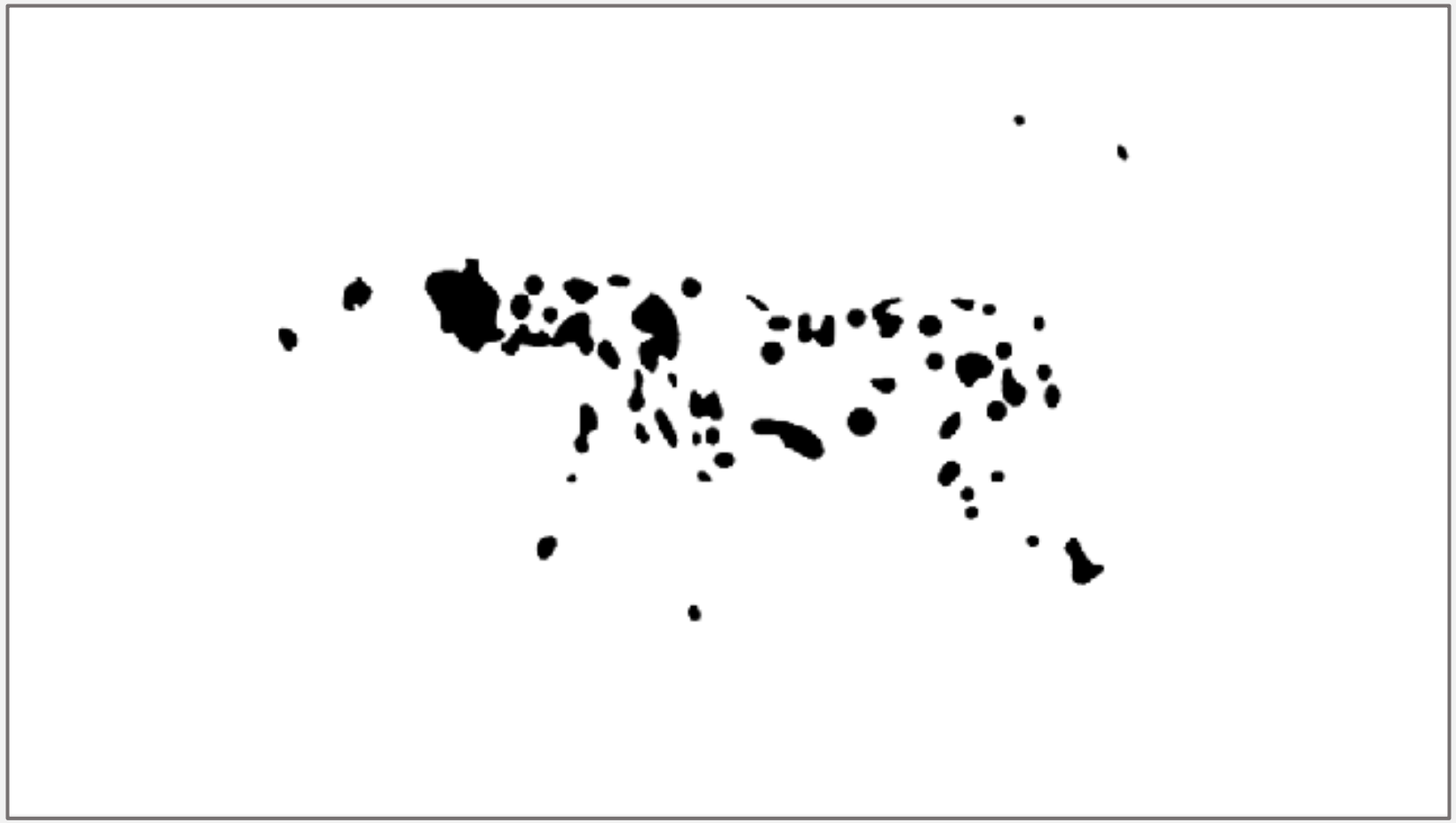


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Gestalt Principles ▶ Closure

(Similarities to Continuity)



Adobe

WWF

usa network

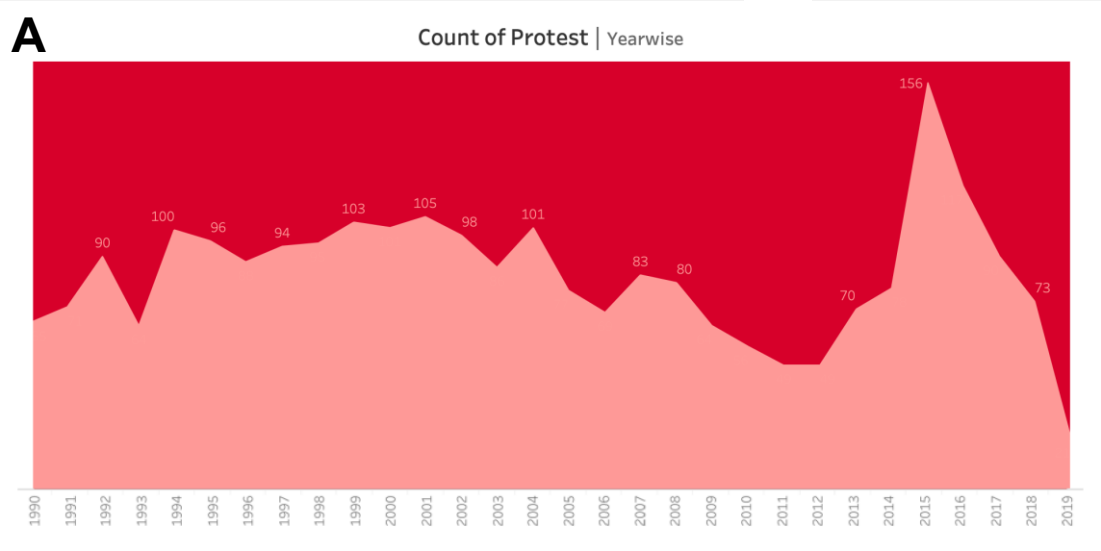
In our effort to recognize patterns, we fill in missing pieces to create a complete image.

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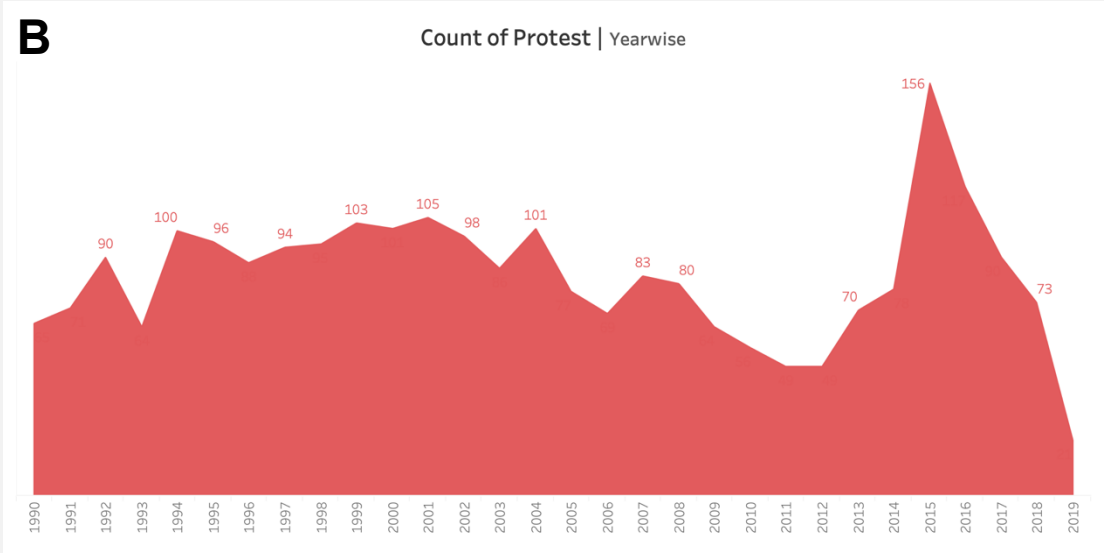
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Gestalt Principles ▶ Figure-Ground

Which chart do you find more readable, A or B?



The low contrast requires extra cognitive load.



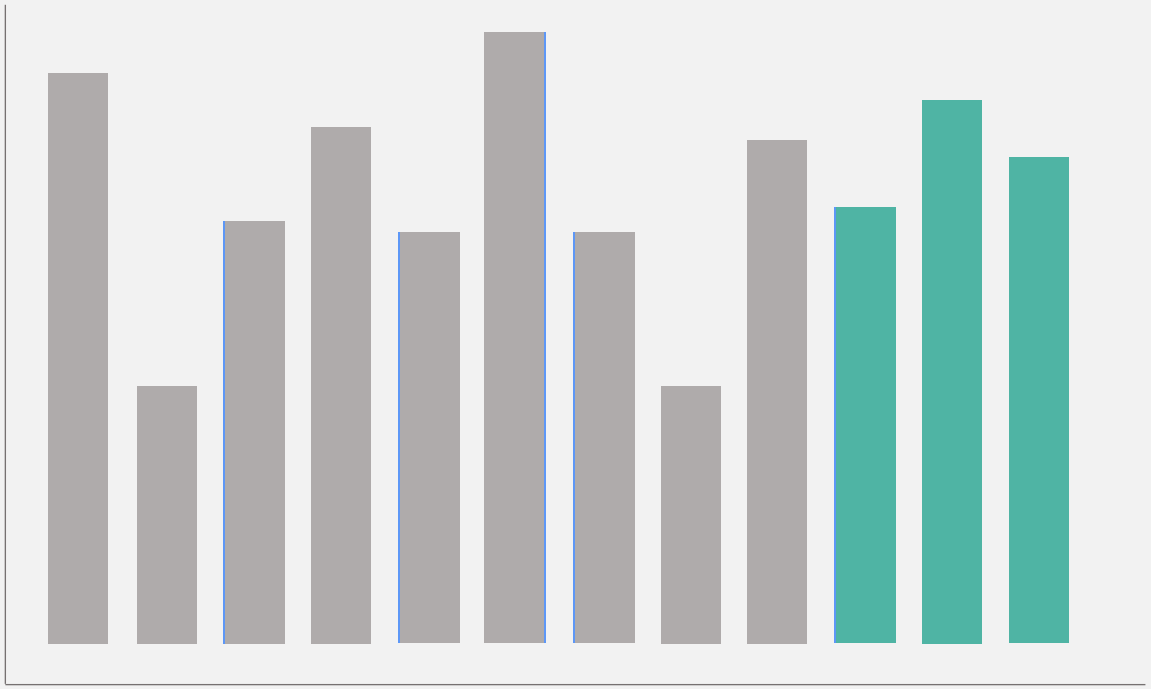
Increasing the contrast improves readability.

We instinctively perceive objects to either be in the foreground (prominent) or the background. It lets us know what we should be focusing on and what we can safely ignore.

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Gestalt Principles ▶ Focal Point



Whatever stands out visually is given priority attention by the audience.

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Gestalt Principles ▶ Order & Symmetry





Not quite the same?

We tend to look for order in objects, and symmetry is one way we do this - a simple, harmonious rule.

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3

Best Practices & Common Pitfalls

The Do’s & the Don’ts



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10 Methods of Deception

How to Spot Lies, Avoid Common Mistakes & Build Effective Data Visualizations



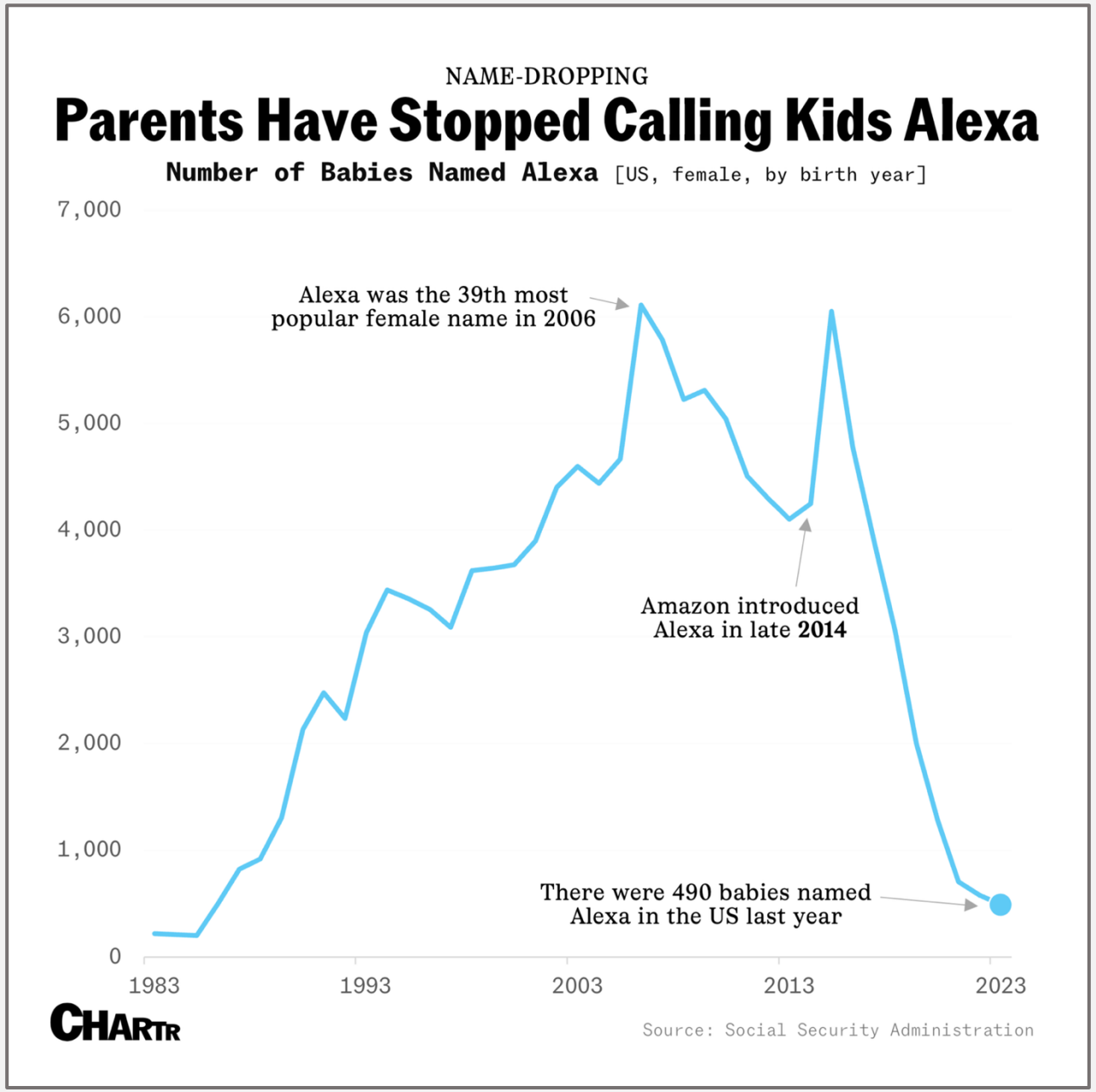
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Simple is Best, Less is More

It's not necessary to overcomplicate a graph when a simple line chart or bar chart is sufficient!

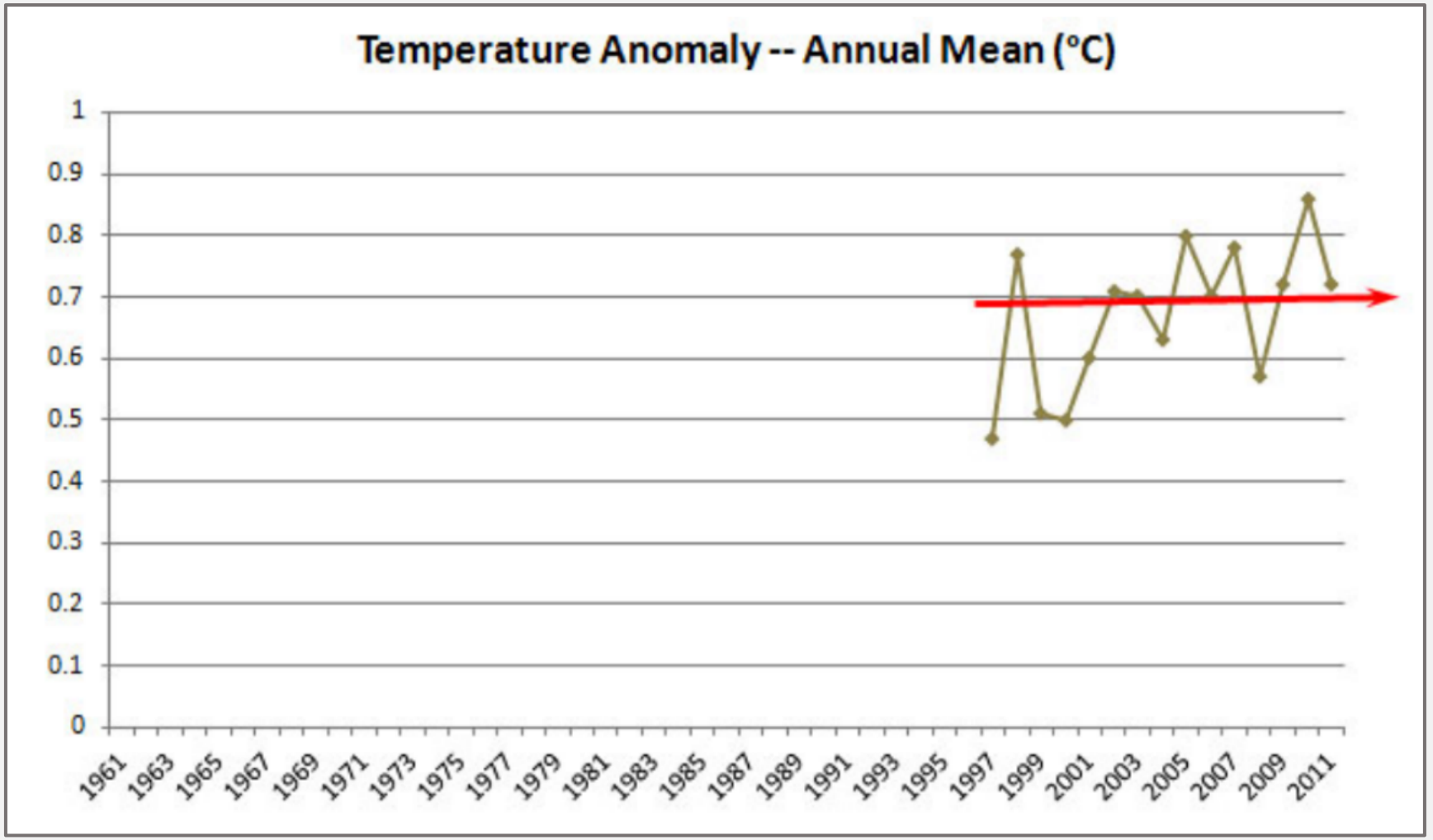
BONUS:
Use annotations (labels) to give context!



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Limited Scope – Tell the Truth

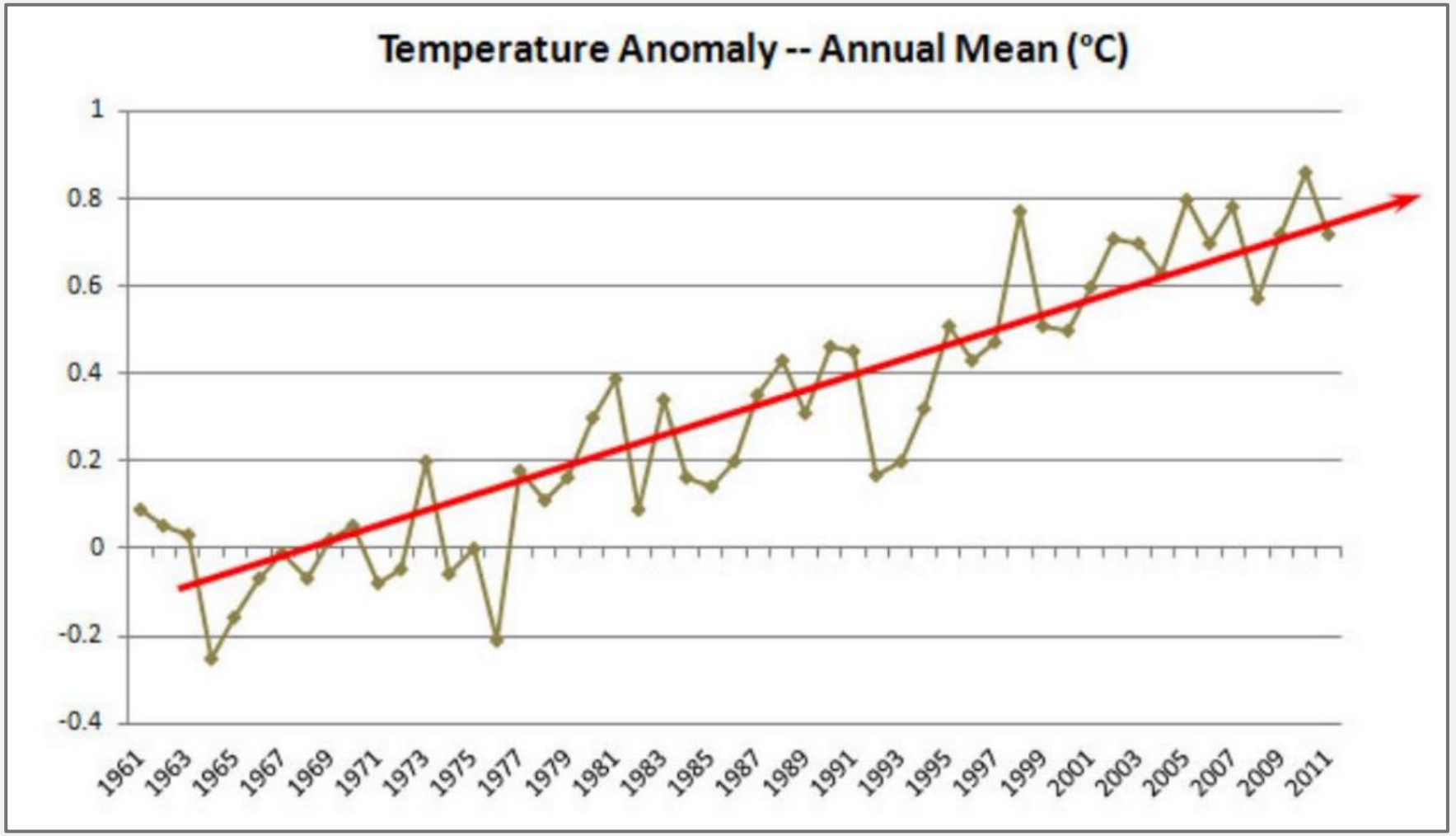


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Limited Scope – Tell the Truth

Avoid distortion!



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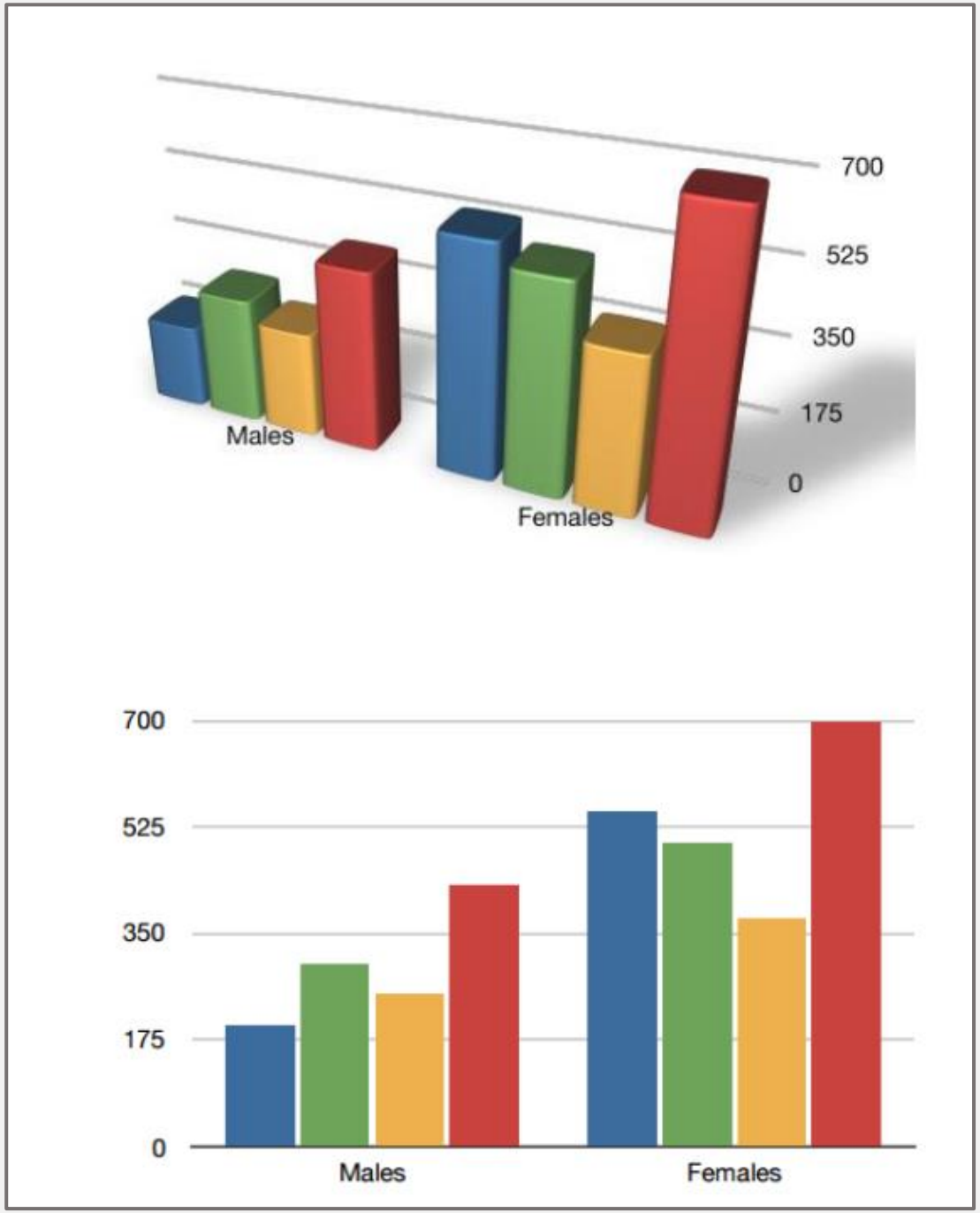
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Maximize Data-Ink Ratio

$$\frac{\text{Data Ink}}{\text{Total Ink Used}}$$

Data ink is the critical part of a data viz.

If removed, the data viz would not make sense.



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Maximize Data-Ink Ratio

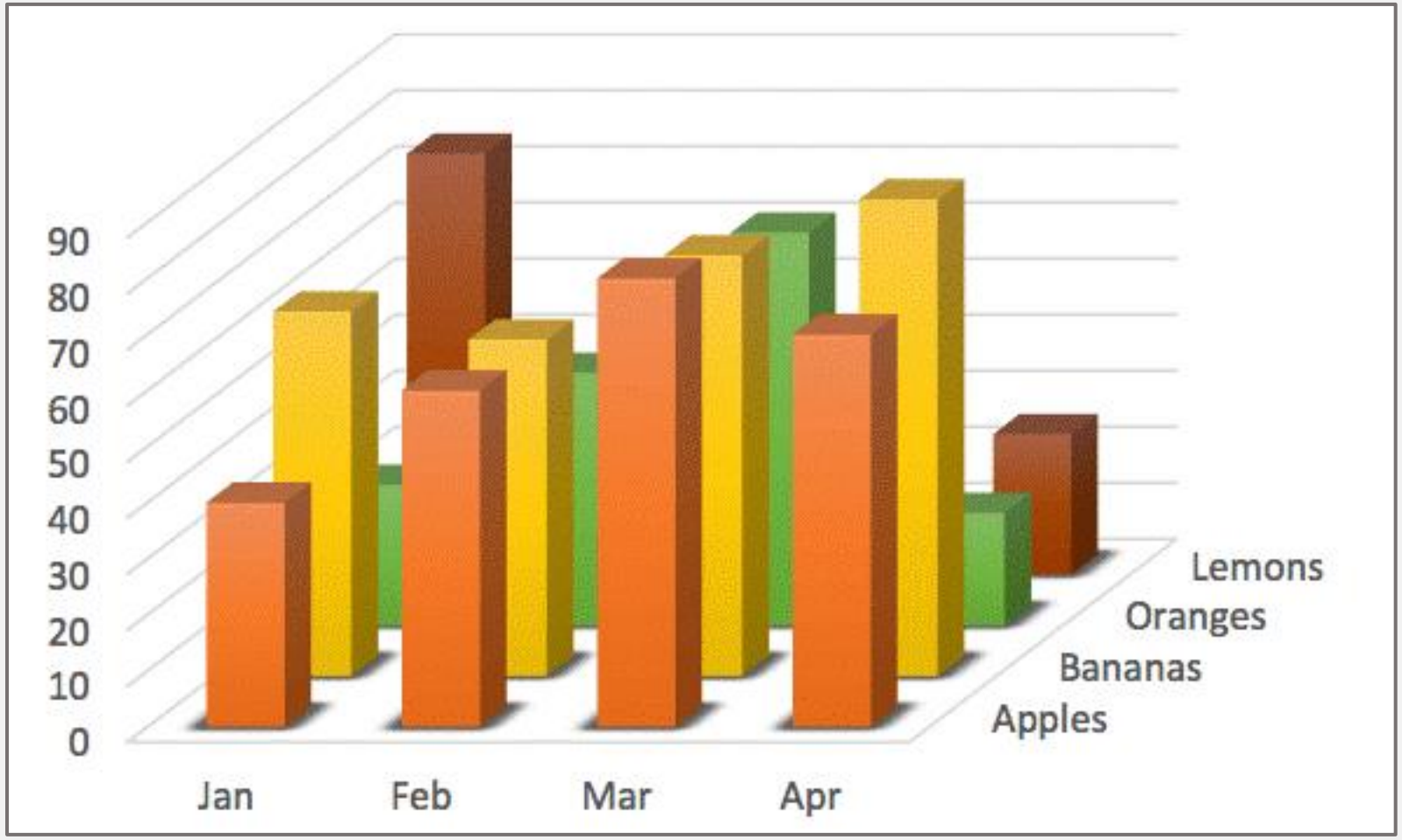


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Maximize Data-Ink Ratio

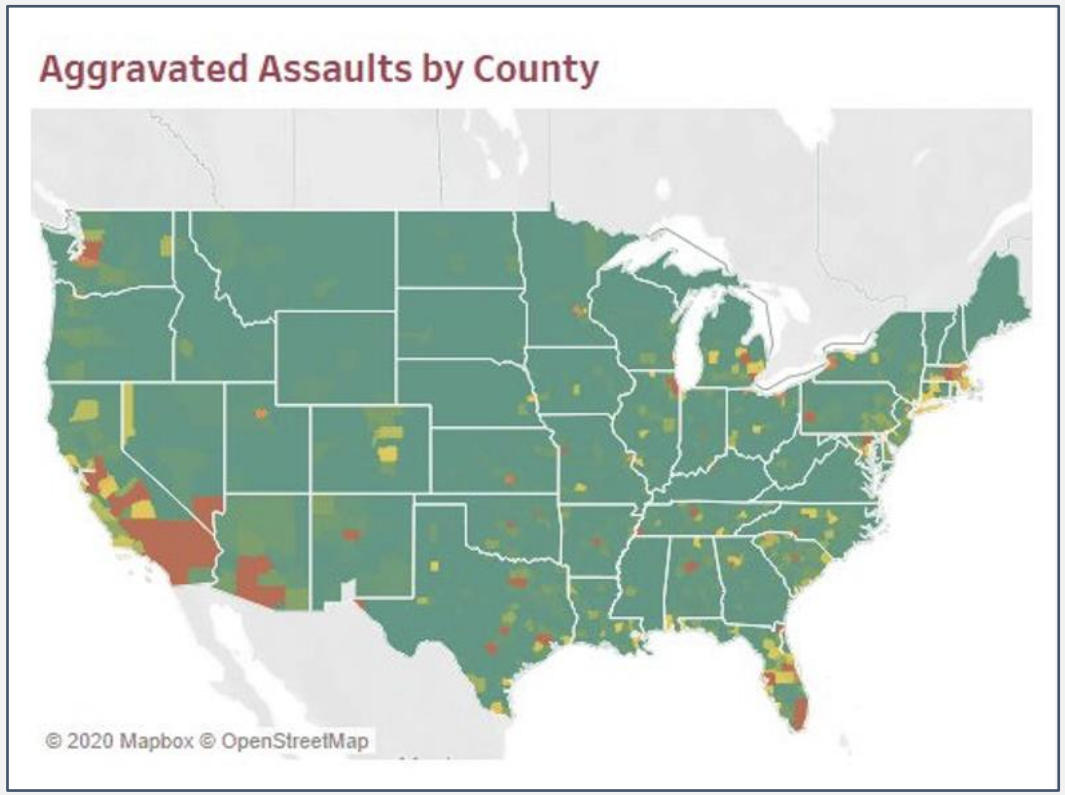
How can we improve this data viz?

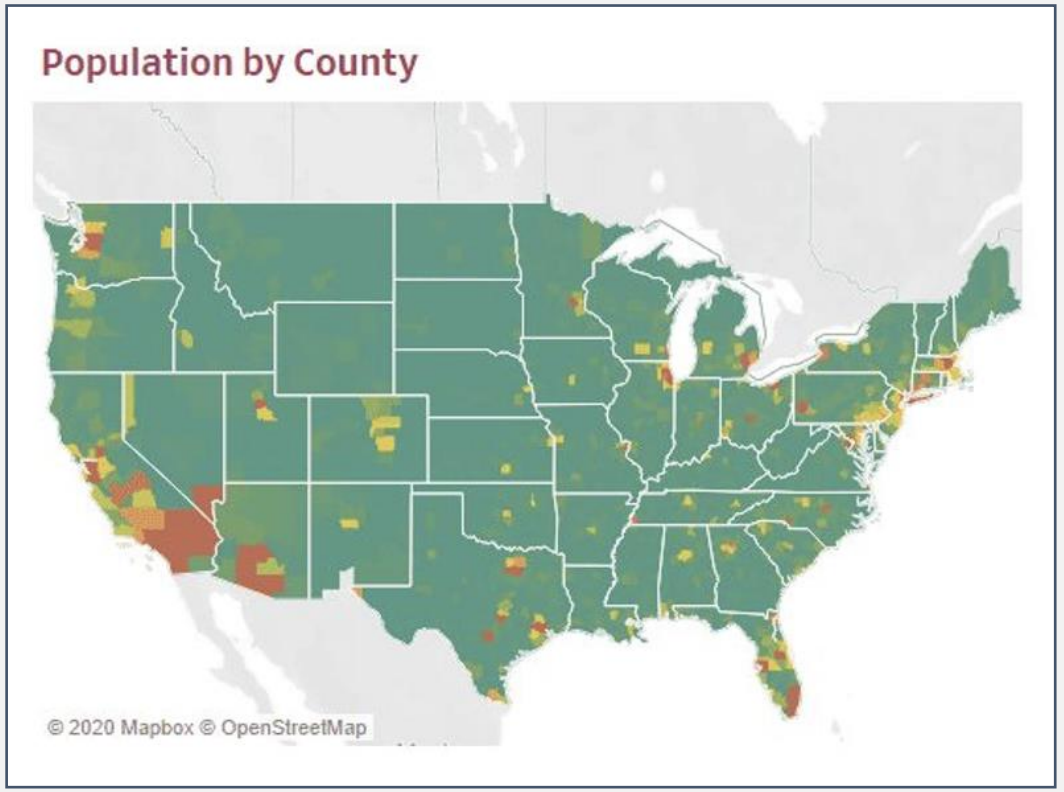


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Using Only Absolutes





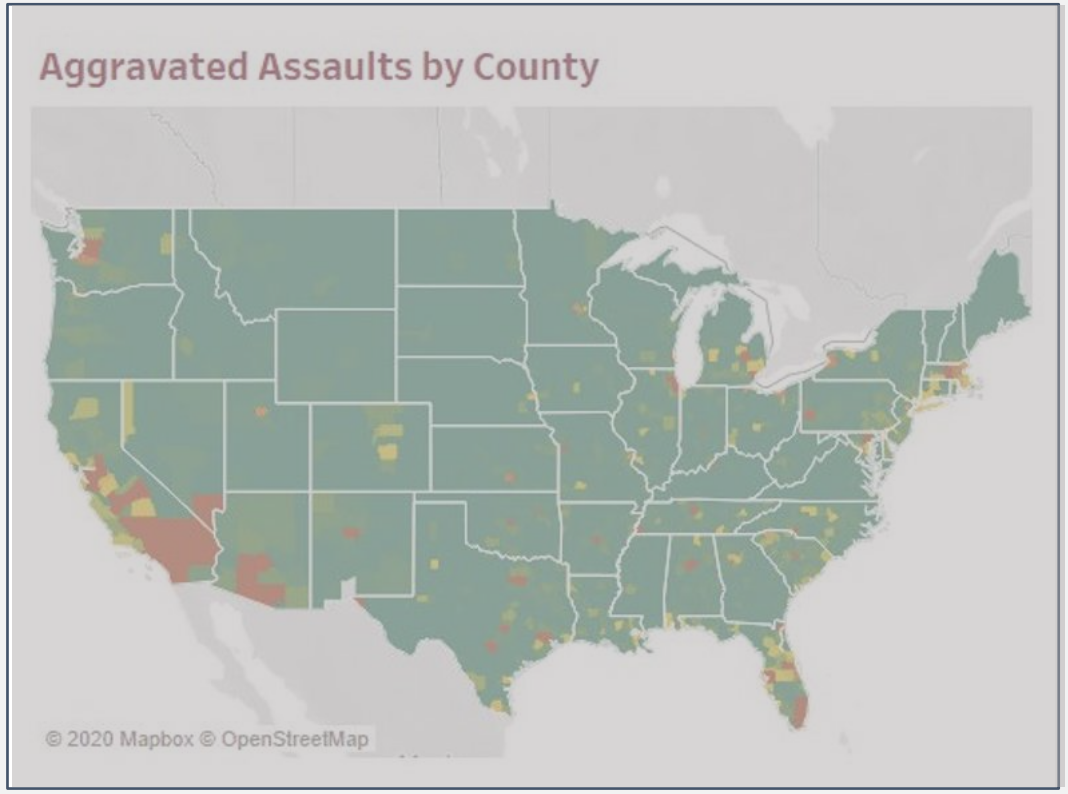
What's wrong with this?

This is just a reflection of population.

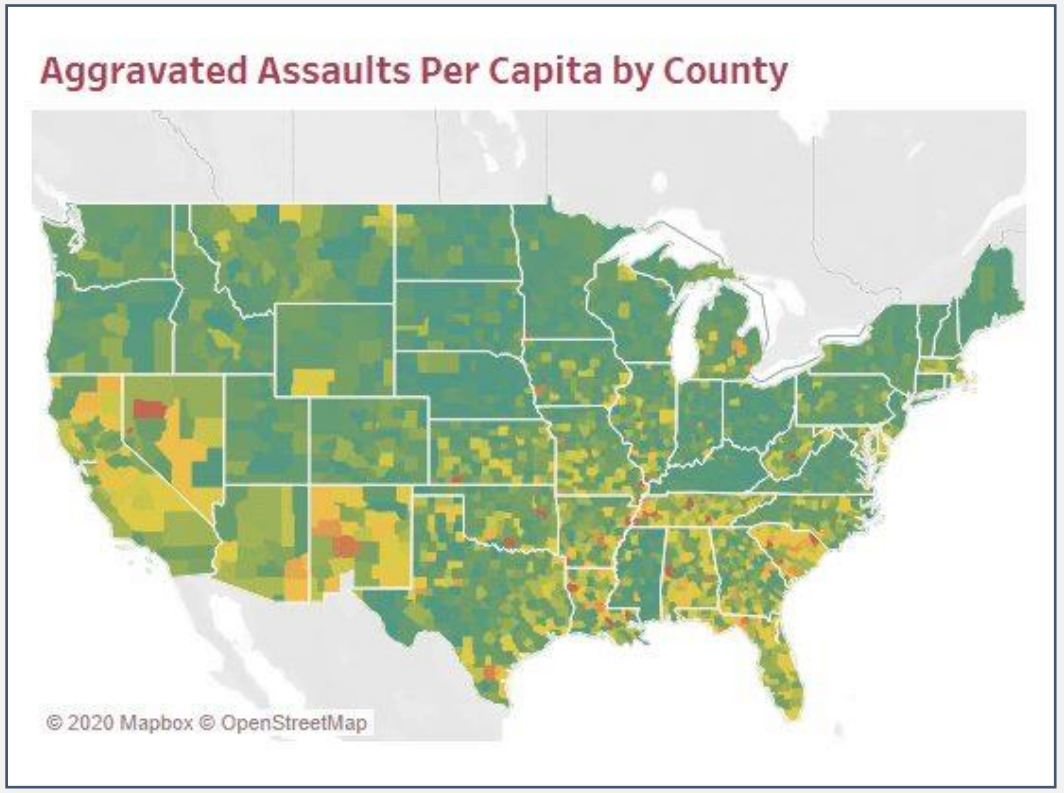
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Using Only Absolutes



What's wrong with this?

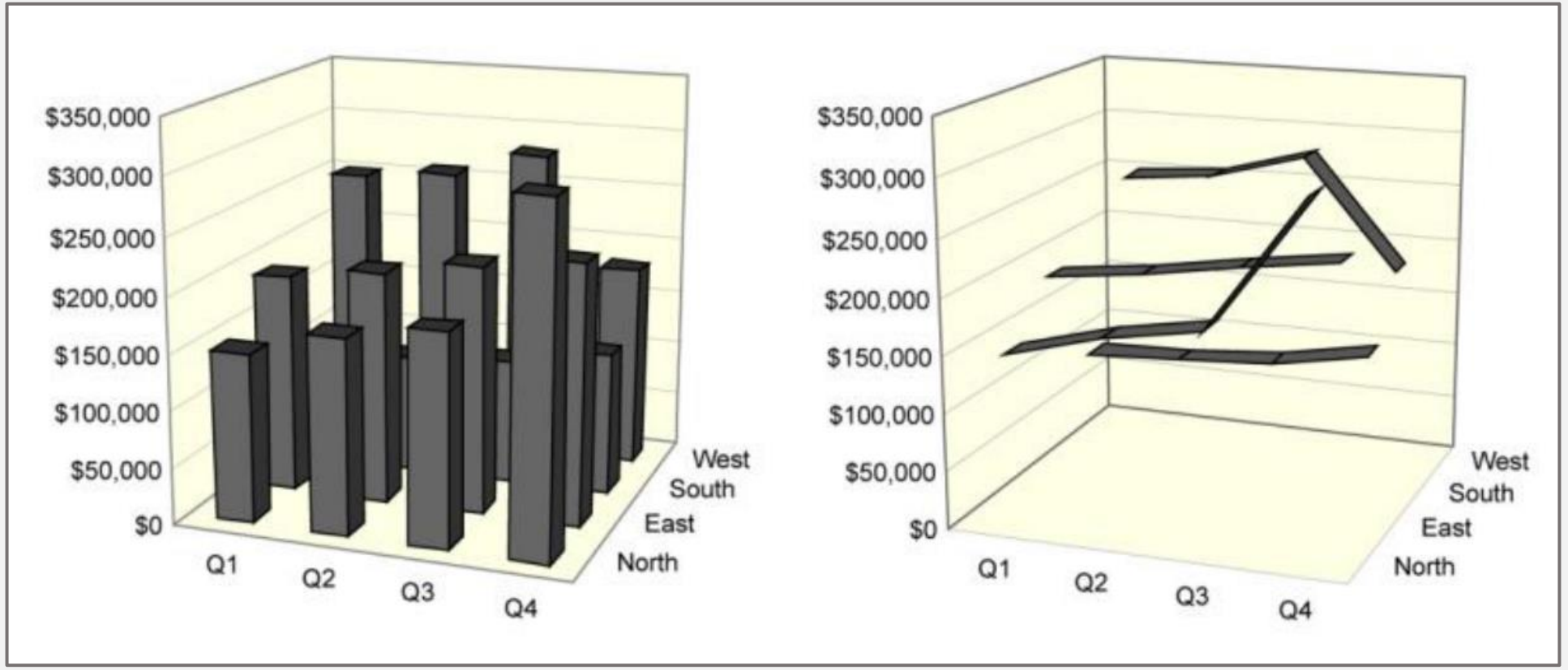


MUCH better! When comparing across places, compare relative values *per capita*.

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Extra Dimensions for No Reason

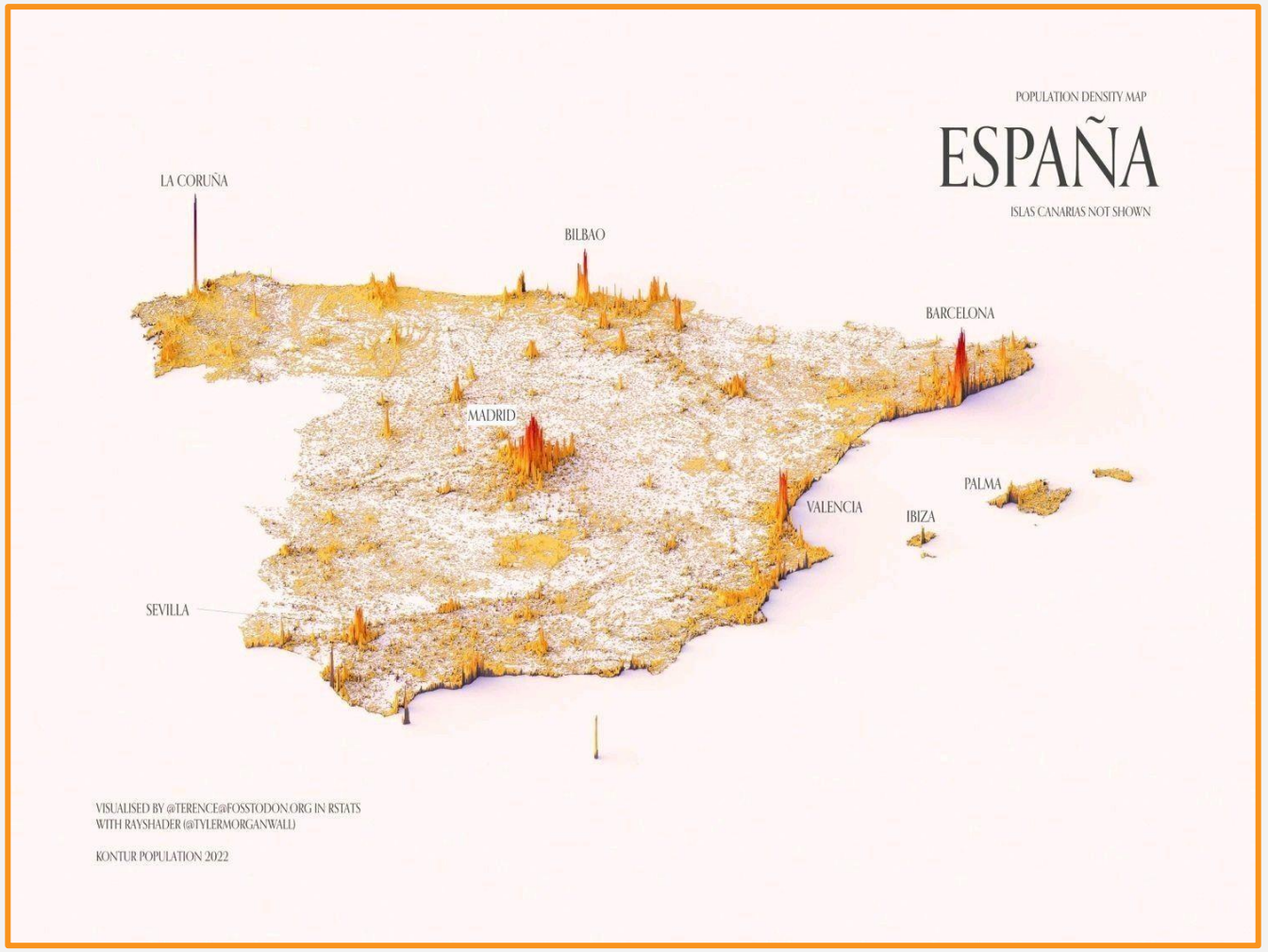


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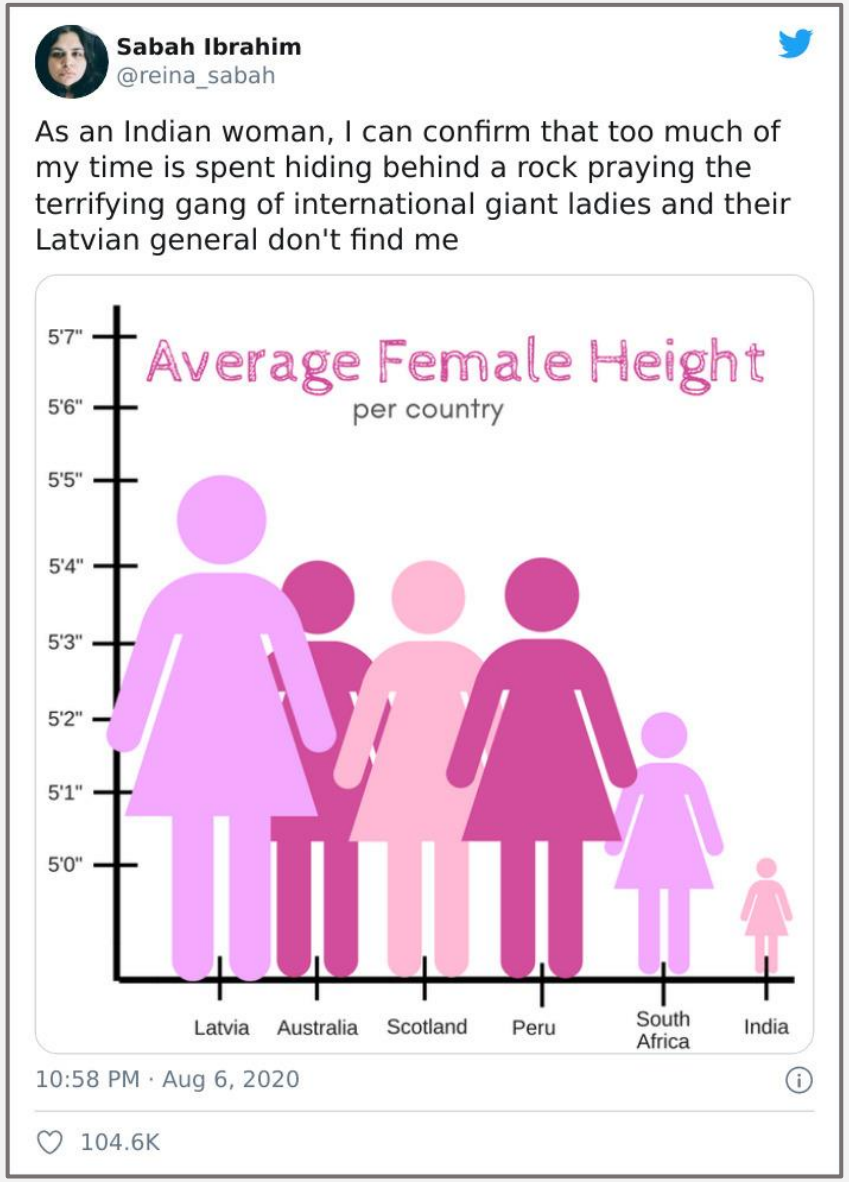
Extra Dimensions for No Reason

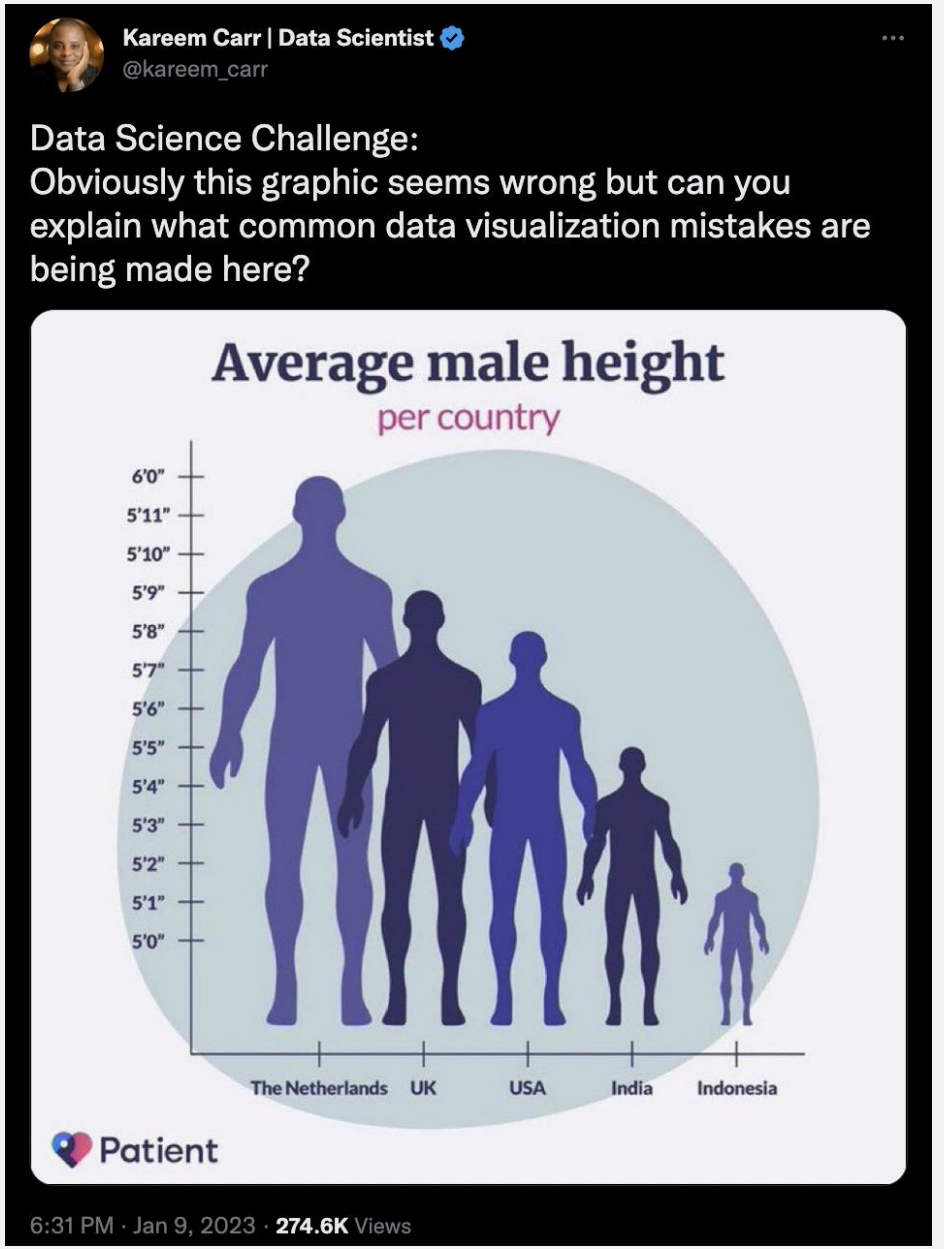
3D is okay *sometimes*, usually geospatial applications



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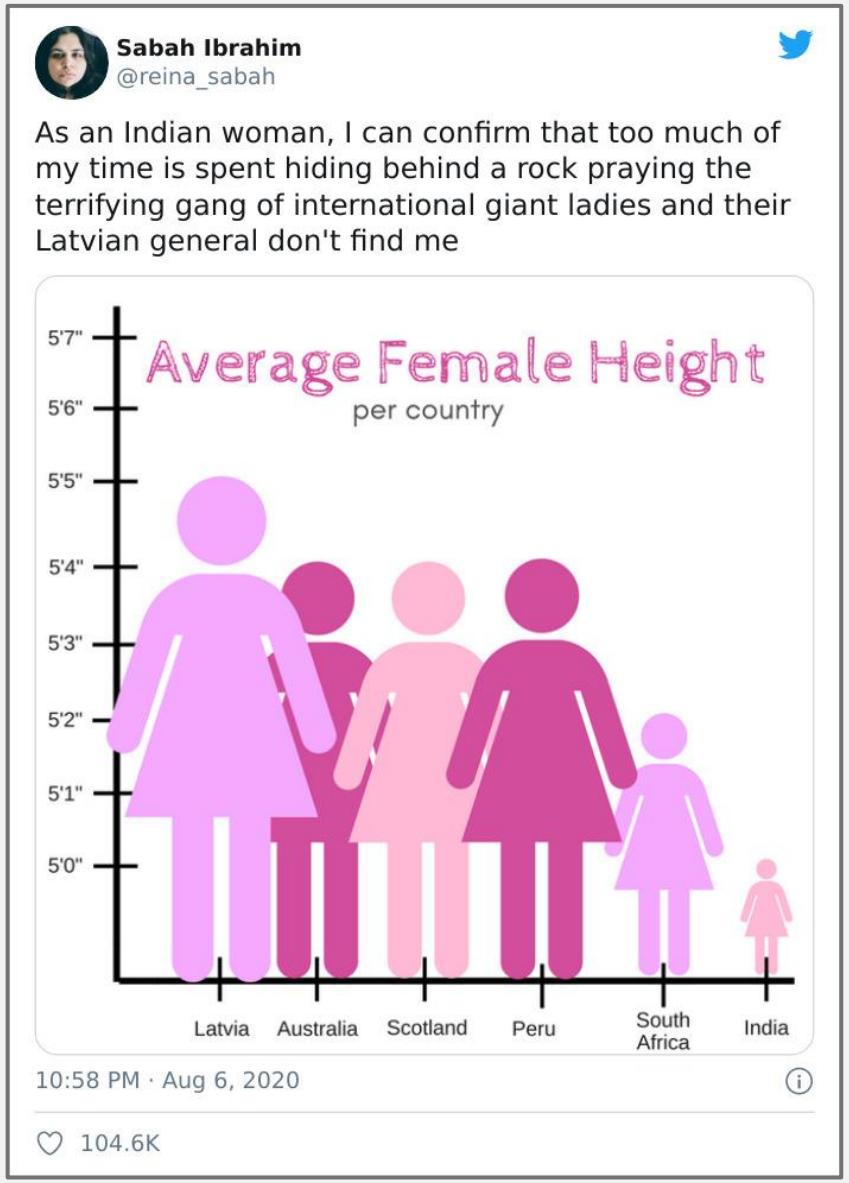


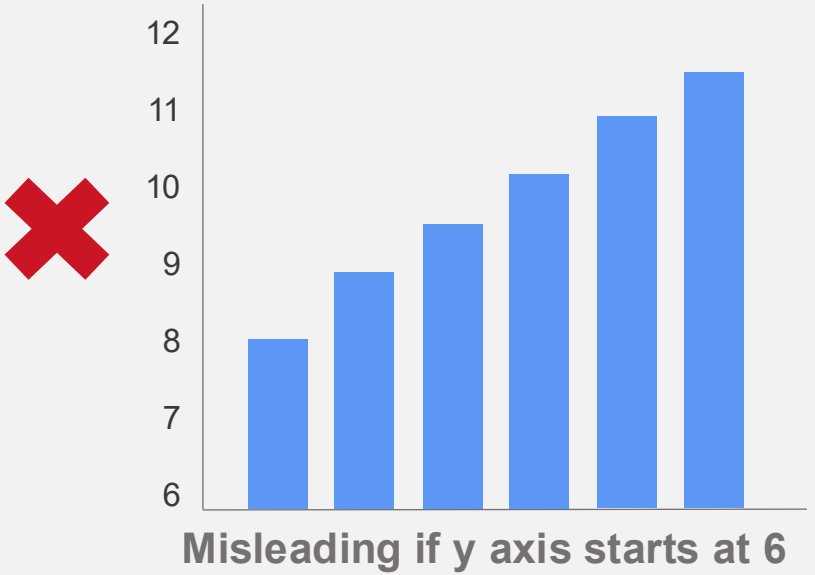


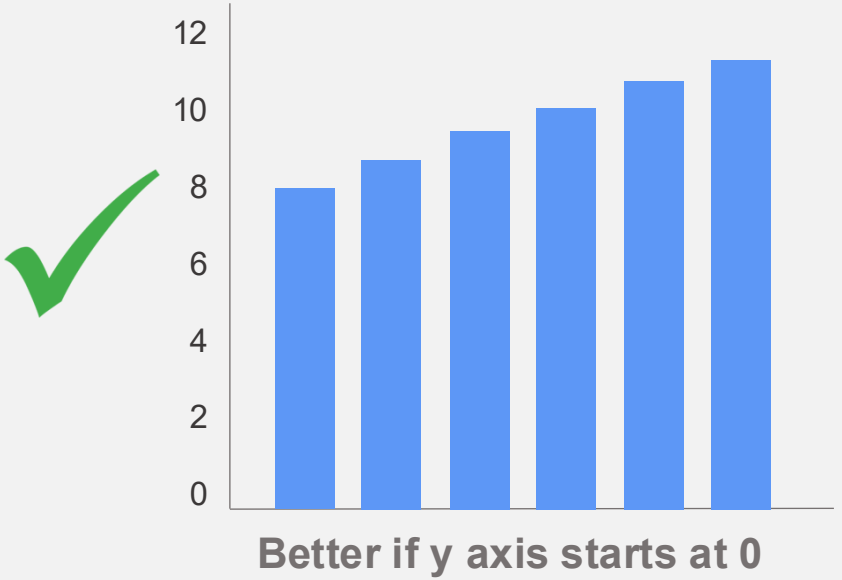
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Truncated Axis



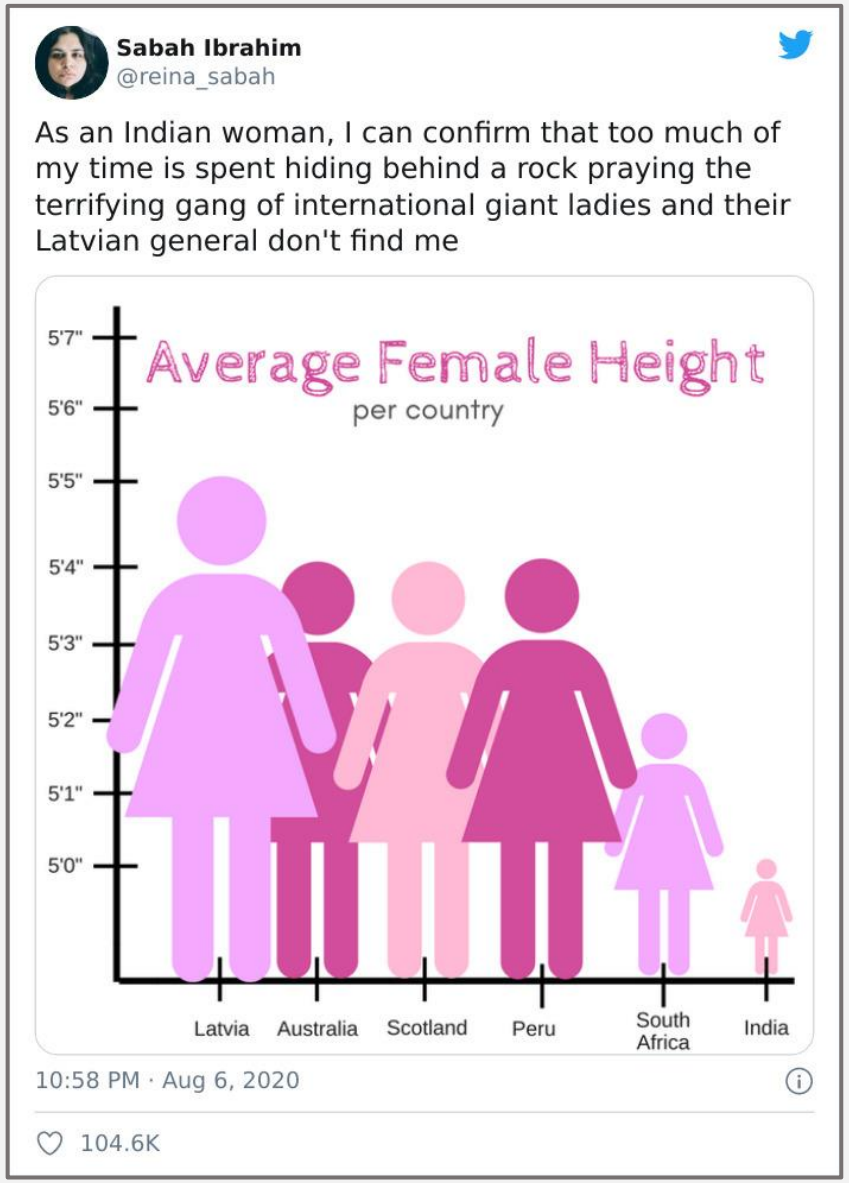


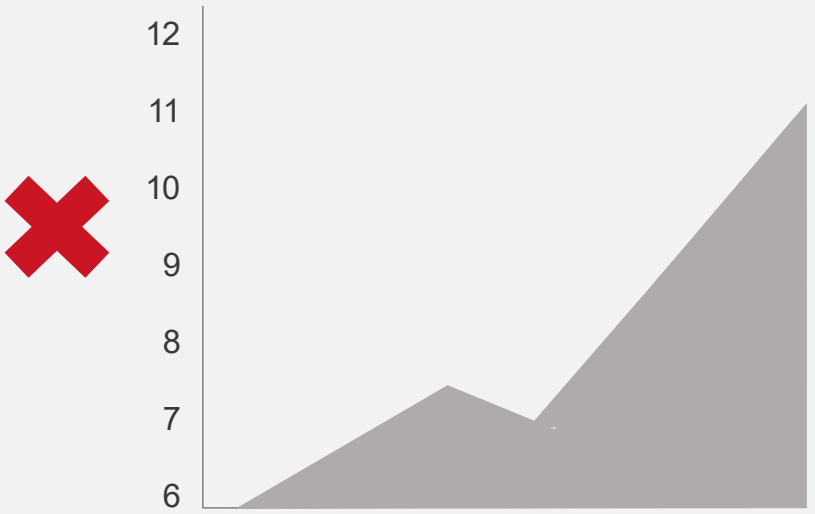


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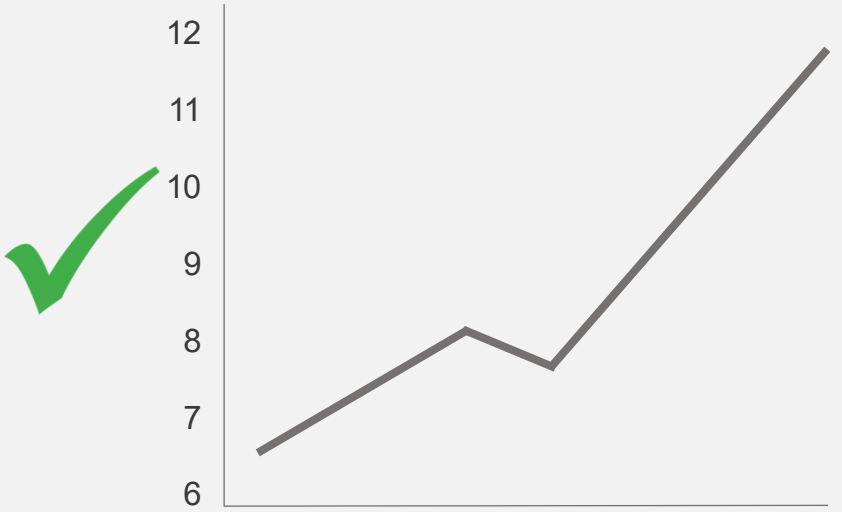
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Truncated Axis





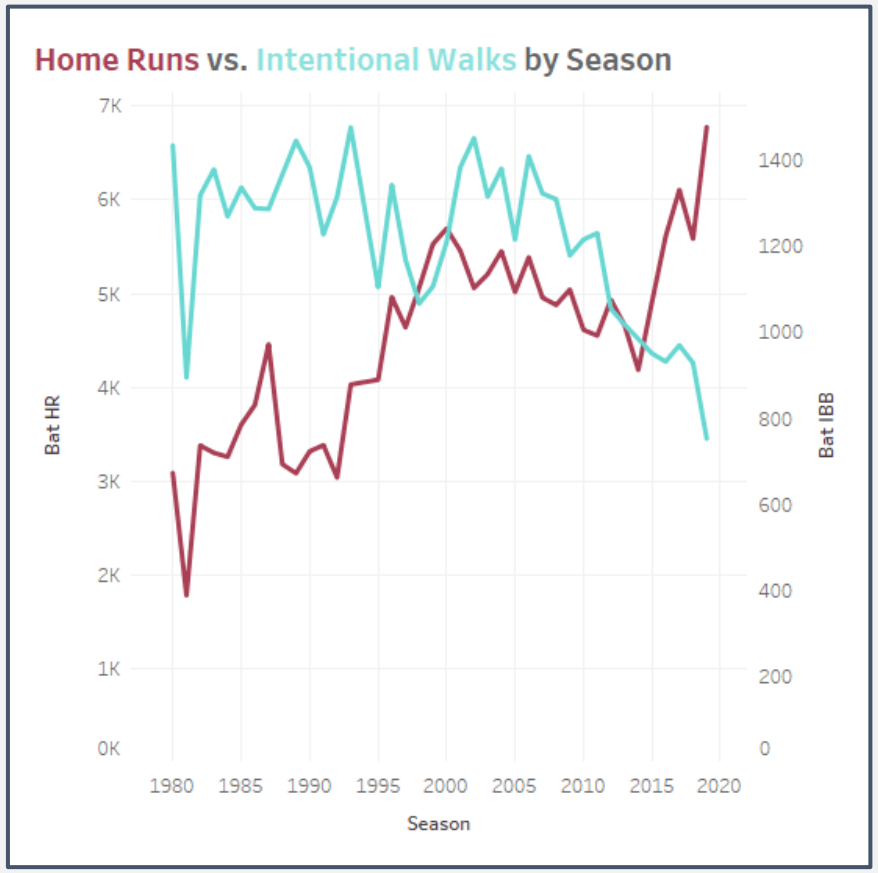
If data point touches floor of chart, do not truncate!

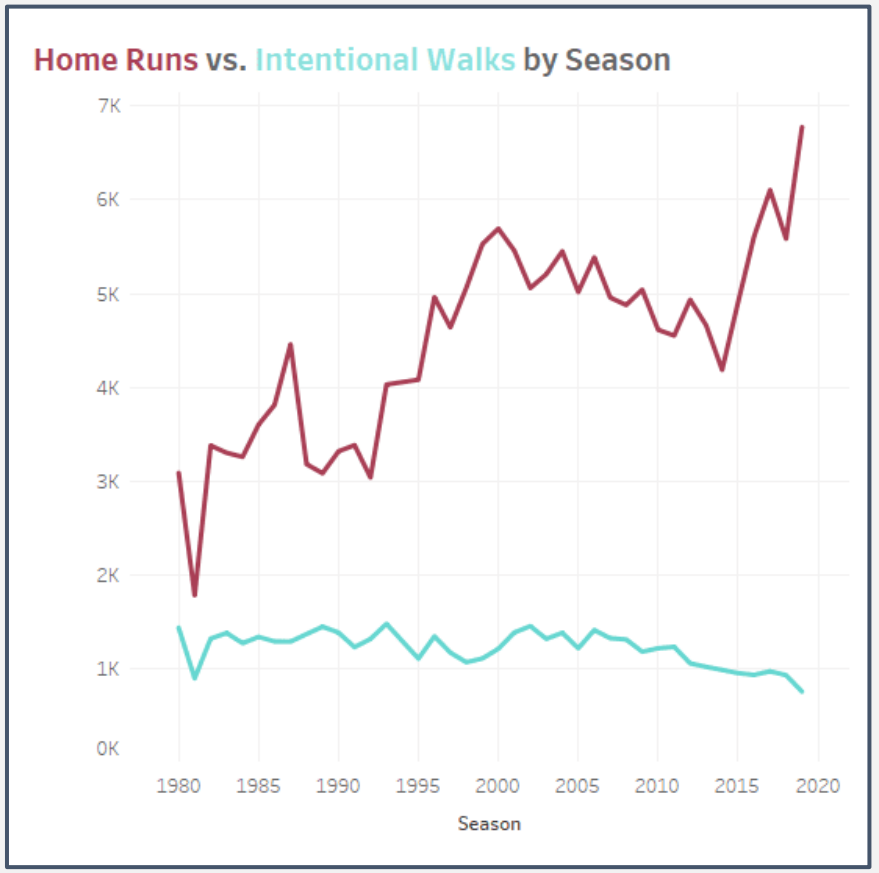


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Dual Axis





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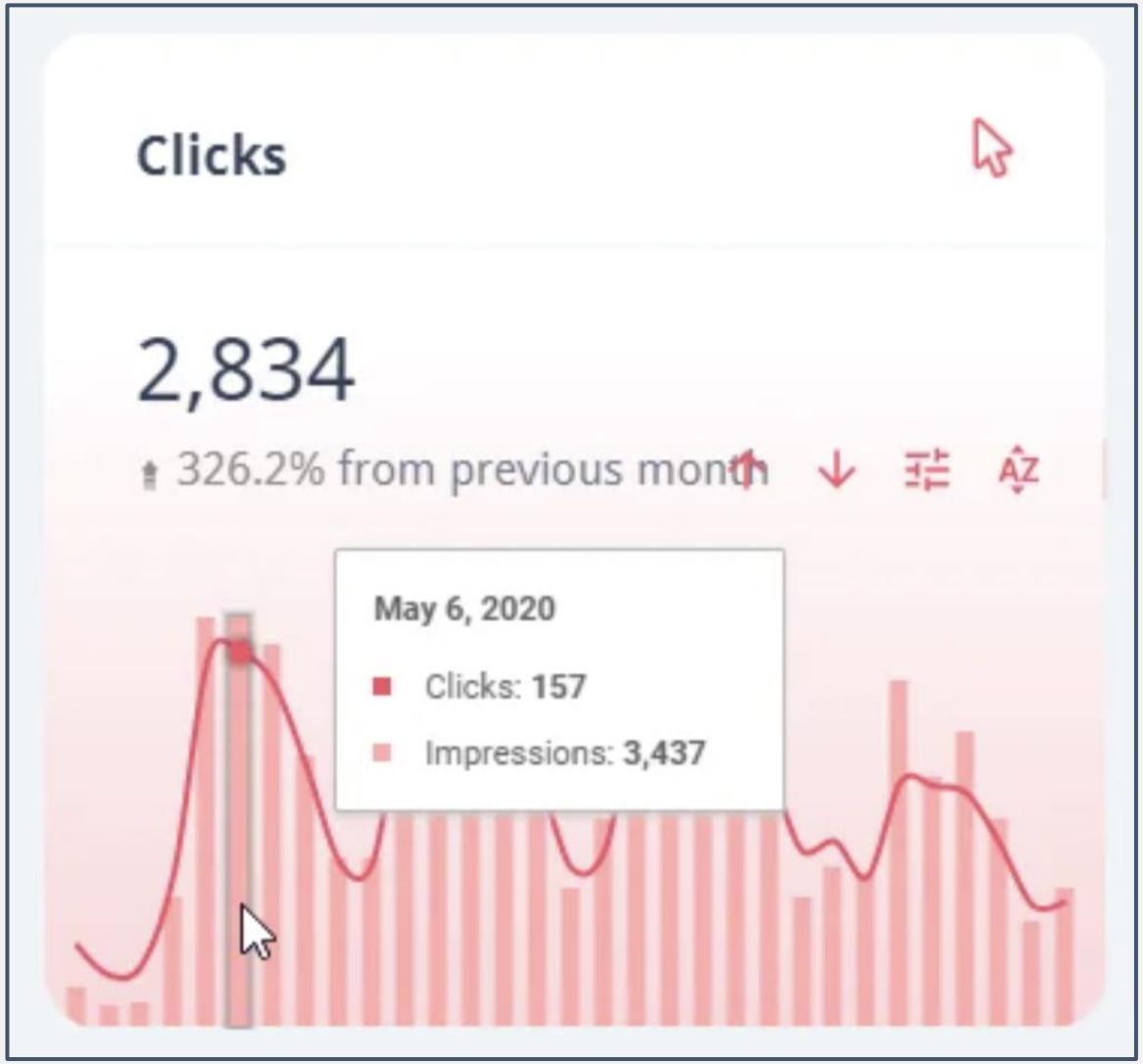
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Dual Axis

This example is okay though!

1. Visualizes 2 variables that have very different scales.

2. Graphs 1 variable with a line and 1 variable with bars.

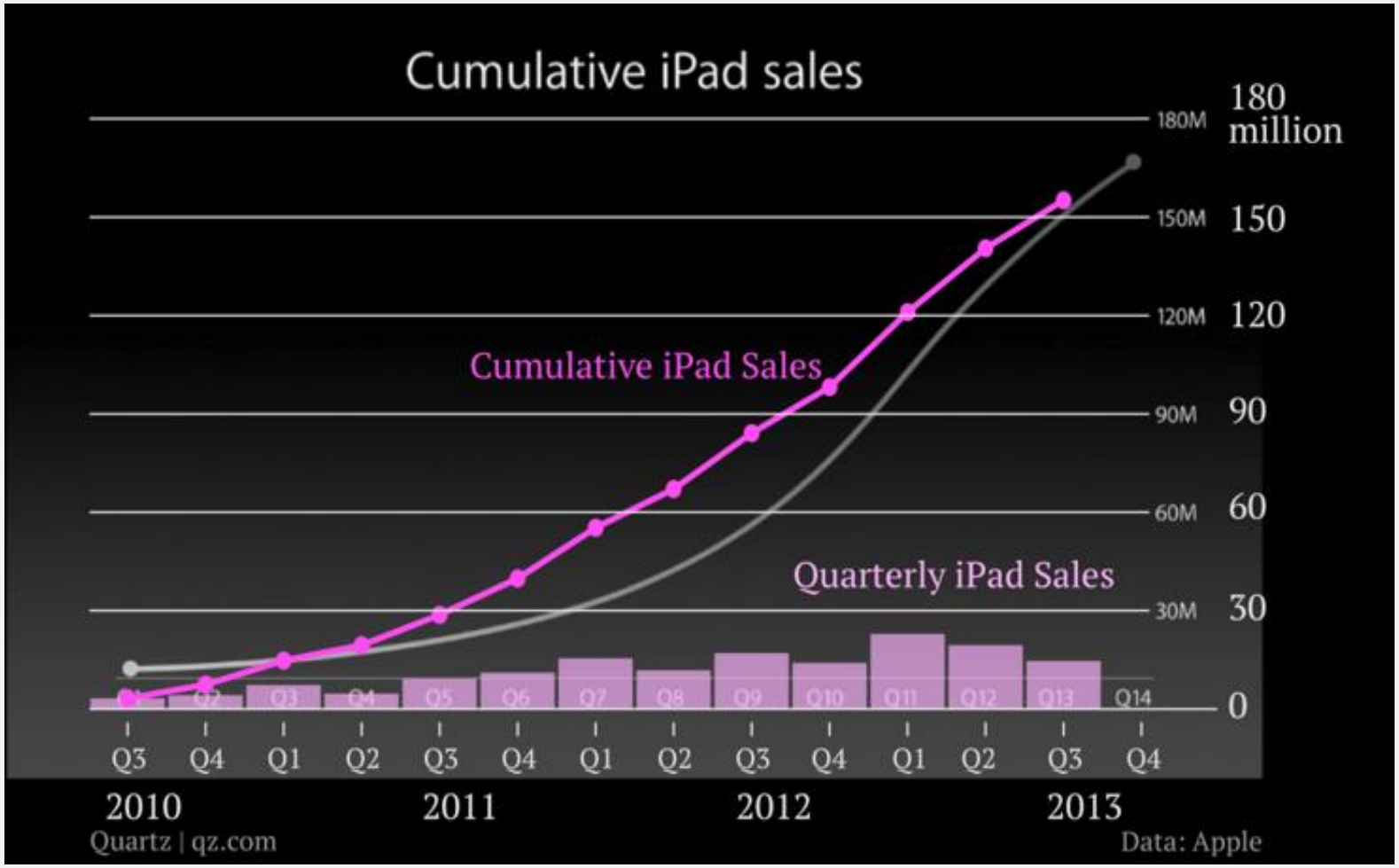


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Hiding Trends in Cumulative Charts

Notice anything that could be deceptive with this?



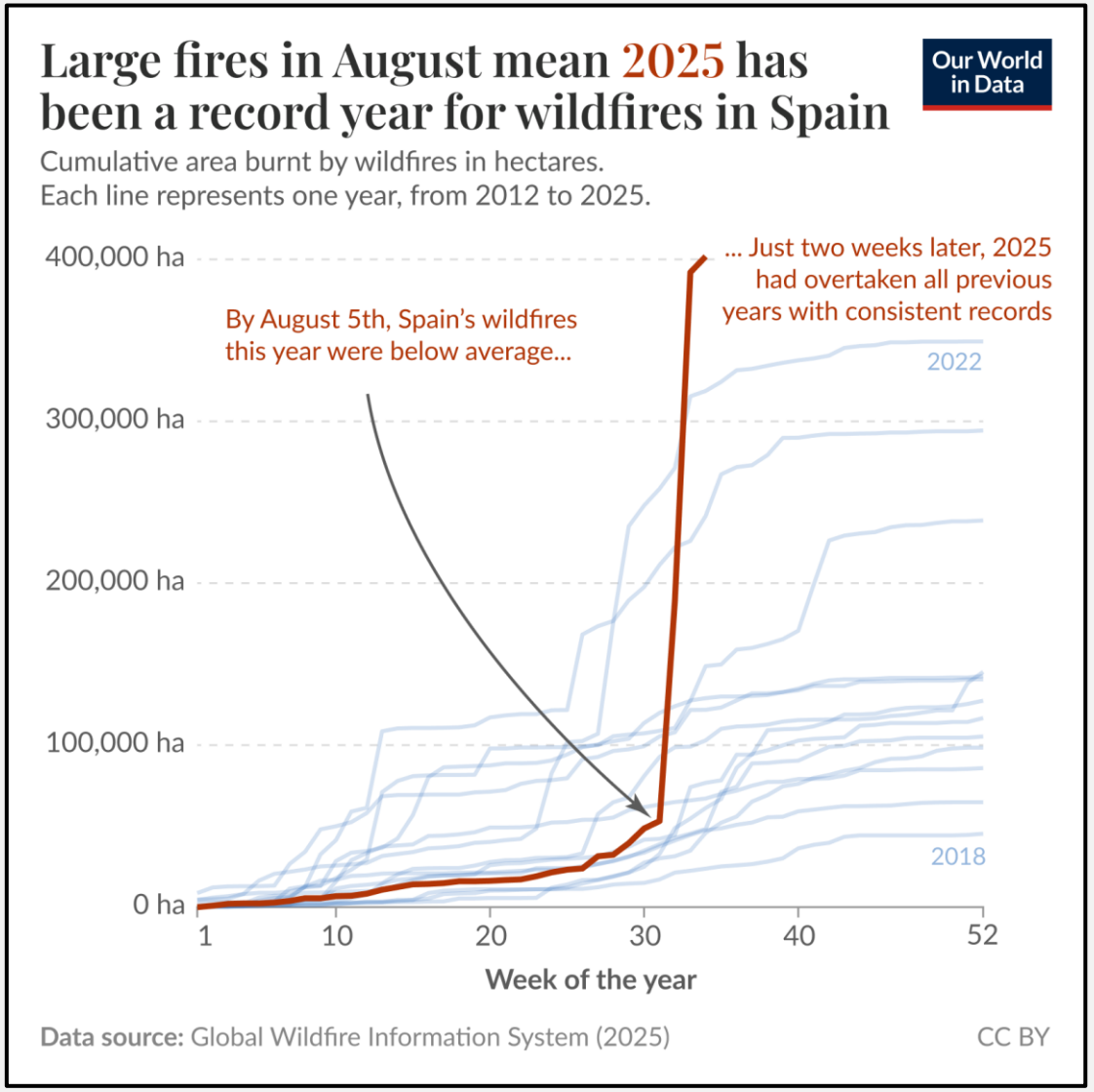


How Apple exaggerated sales of its iPad: Chart shown at launch is 'misleading' because it fails to show recent dip in sales

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Hiding Trends in Cumulative Charts

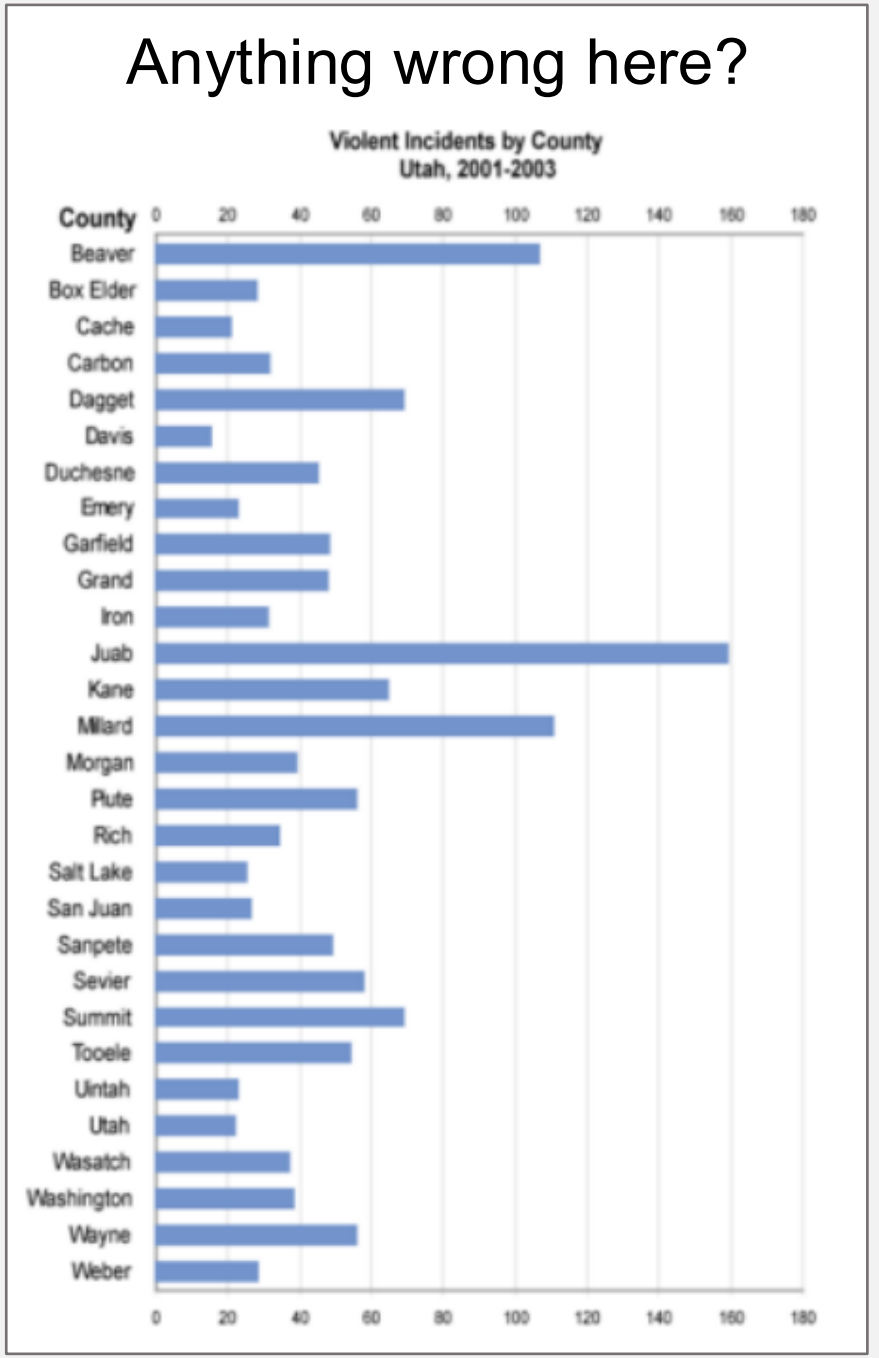


You can use cumulative charts, but only if you are sure they **make sense for the story** you want to tell and that your **audience understands** how to use them.

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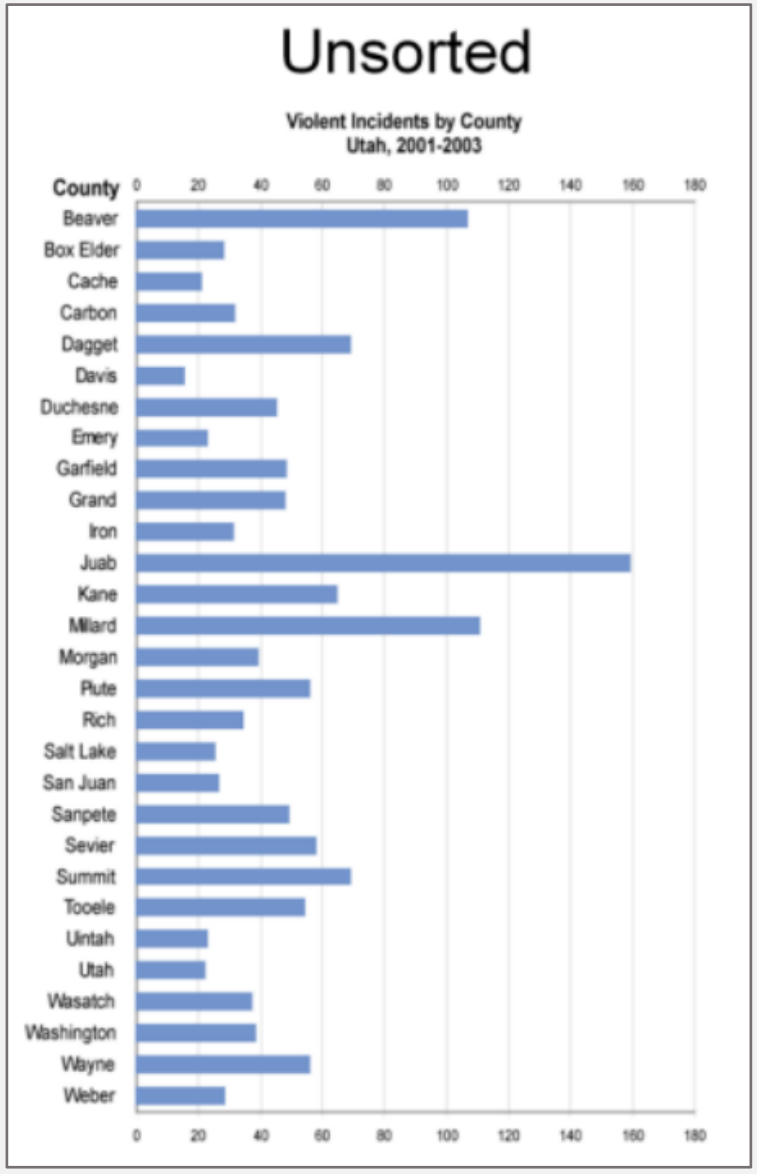
Anything wrong here?

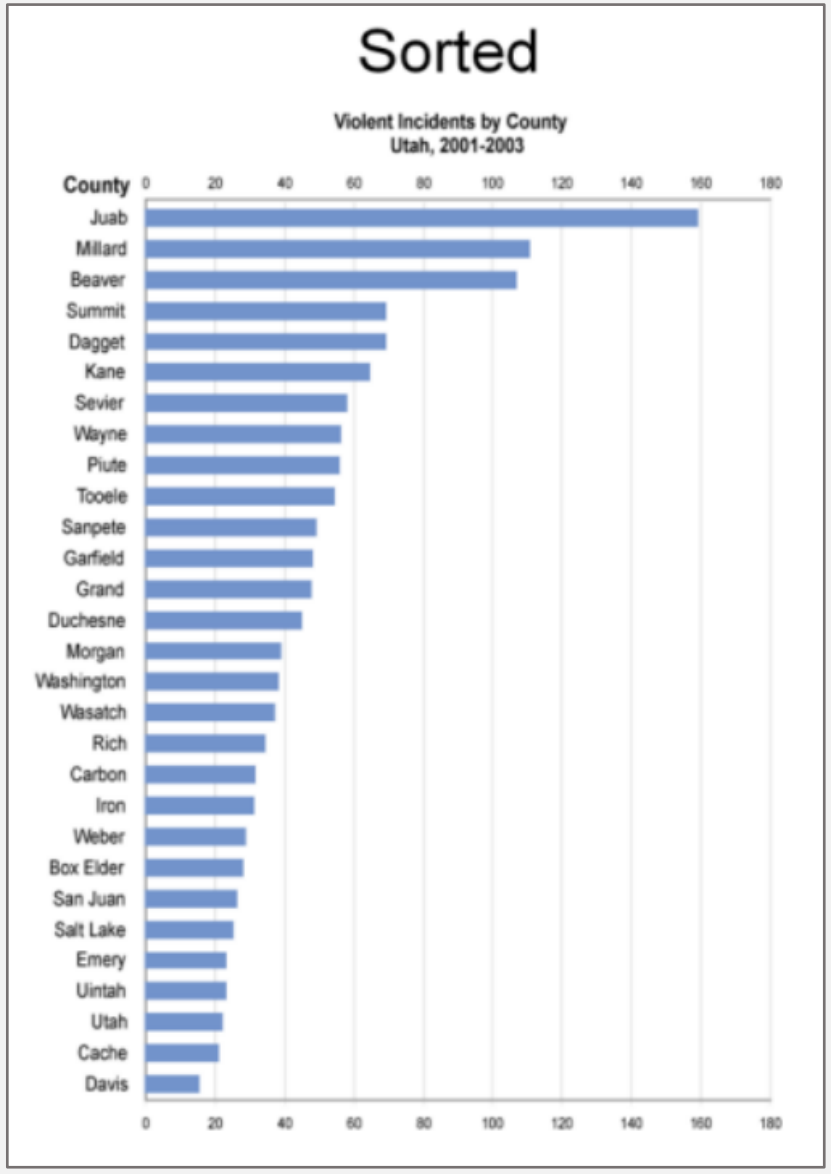


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Keeping Order





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And you will read this last

You will read this first

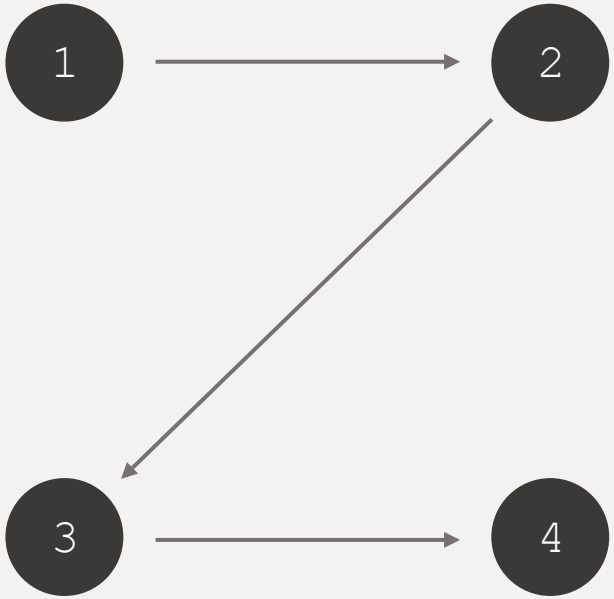
And then you will read this

Then this

Keeping Order

Take advantage of the way the mind works when building text or a visual

Remember the **Z pattern** when designing a dashboard.

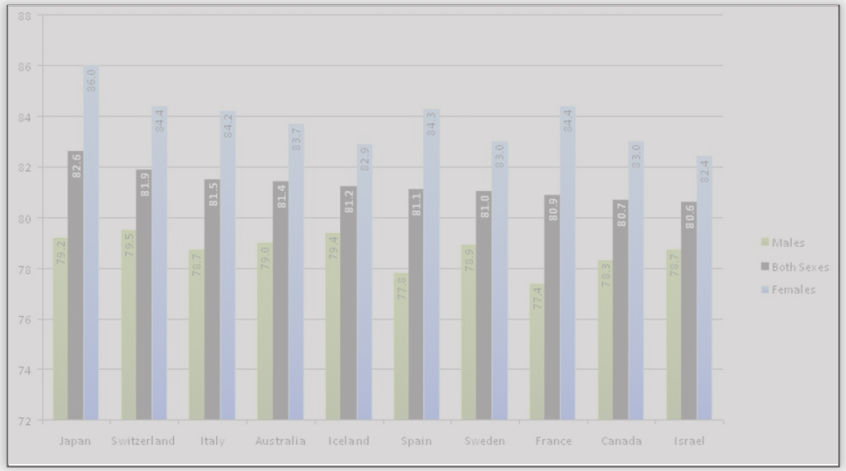


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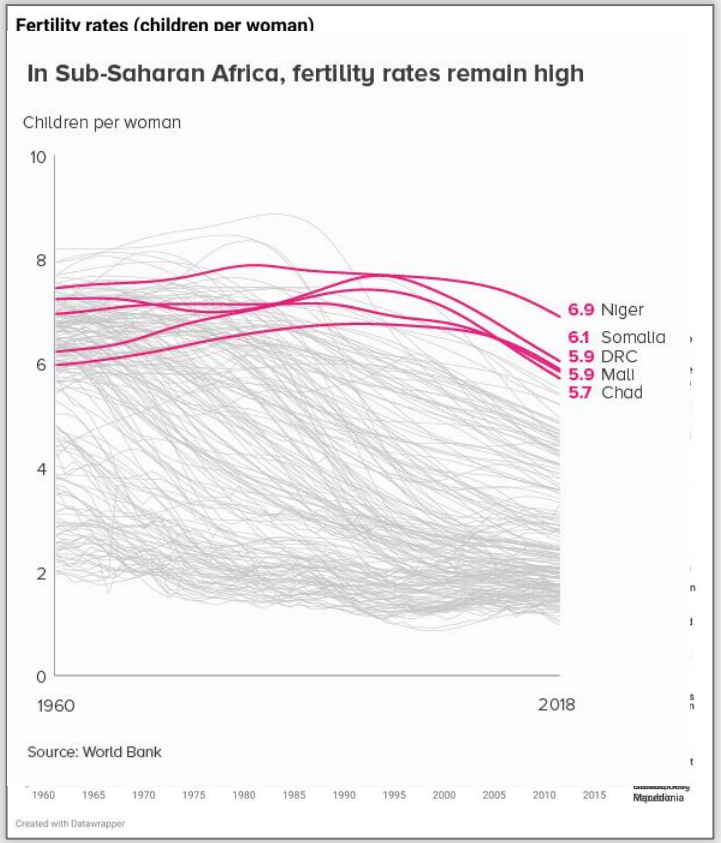
Remember Thresholds

< 10 bars



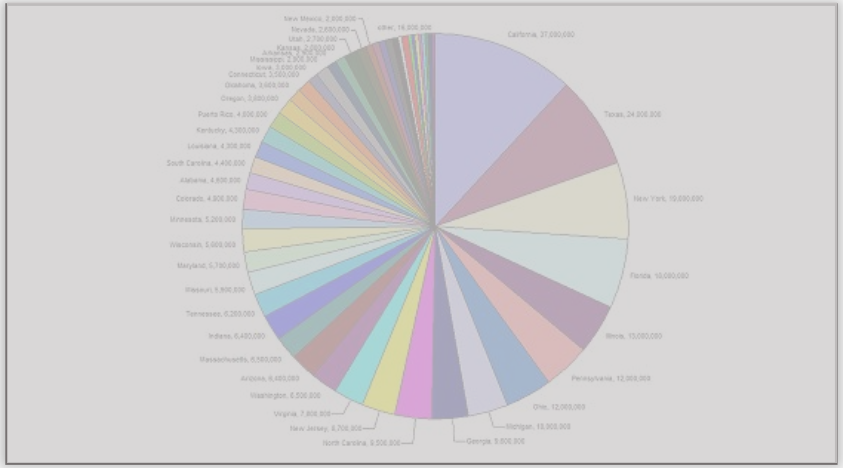
Filter or group more
Split into separate charts

=< 6 lines



Filter or group more
Split into separate charts

2-5 slices



Filter or group more
Add an 'Other' category

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Feel free to stay in touch!

(I've reached the 30K connection limit on LinkedIn unfortunately, so until I can get that back down, you can follow me there!)

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